Anatomy of a Smooth Transition: A Task-Based Macroeconomic Model of AGI Displacement with Fiscal and Social Safety Nets

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Abstract

Artificial general intelligence (AGI) differs from automation in that it enters the creation of new tasks itself, so the engine that historically reinstated displaced labor may collapse endogenously. We build a task-based model with two-layer task structure in which the human reinstatement rate declines as AI capability advances, plus nominal rigidity, the zero lower bound and debt-deflation, capital-tax financing of workfare and transfers, and a social-instability channel that is independent by construction, entering through a separate equation rather than through economic deflation. The high-employment equilibrium reaches and crosses the switching manifold through a persistence-type boundary equilibrium bifurcation of the piecewise-smooth fast subsystem, continuing as a stable low-employment equilibrium on the other side within that frozen subsystem; on a full dynamic path an effective crossing flattens the labor-share decline rather than producing interior convergence; we derive a closed-form critical reinstatement rate, make it endogenous to workfare and retraining, and characterize with geometric singular perturbation theory the finite-rate lag between slow policy and fast employment. Three magnitudes are anchored to U.S. data; the rest are declared settings. Under these settings, the loss-minimizing policy on the reported grid is a workfare-dominant corner financed by a minimal capital tax, whereas no-policy and pure-transfer regimes are infeasible. The minimizer is grid-dependent rather than identified; we report the corner retaining a buffer. At the baseline frontier growth $g_A = 0.010$, consumption stays above the floor $C^{min} = 0.30$ with $\min_{t \geq 10} C_t = 0.357$, and the financed buffer still holds at $g_A = 0.05$ ($\min_t C_t = 0.344$); the floor is breached only at $g_A = 0.10$ ($\min_t C_t = 0.294$), after which

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the path crashes, whereas fiscal parameters are nearly inert. Frontier-growth shocks (σ_ε up to 0.03) produce no economic crash across 200 paths of the reported corner, and the fifth percentile of minimum consumption stays at 0.354, although a fraction of paths no longer reach the employment target. The model-implied message: a consumption-smooth transition, in which persistent deflation is nested rather than independently vetoed, requires a financed buffer whose sufficiency is conditional on frontier speed. *JEL classification:* E24, E31, E62, J64, O33.

Keywords: Artificial general intelligence, Task-based model, Boundary equilibrium bifurcation, Geometric singular perturbation theory, Public employment, Labor share

1. Introduction

Artificial general intelligence (AGI) and large-scale automation portend a fundamental reallocation of work. Task-based analyses of recent automation show that new-task creation has historically reinstated labor displaced by machinery (Acemoglu–Restrepo) [2], but the AGI setting is different in kind: AGI participates in *new-task creation itself*, so the reinstatement engine may collapse endogenously. The recent AGI macroeconomics literature splits into two poles. On the *supply side*, Restrepo (2025) [29] and Korinek–Suh (2024) [22] characterize conditions under which automation collapses wages: when the complexity of tasks humans can perform is bounded and full automation is reached, or when automation outruns capital accumulation. On the *demand side*, Fornaro–Wolf (2026) [15] stress that automation shifts income from high-MPC workers to savers, weakening aggregate demand and lowering the natural rate. Yet within this recent literature the two poles are typically treated separately, and the wage-collapse question is not, to our knowledge, joined to the *transitional fiscal policy* needed to sustain demand while the labor share vanishes.

This paper proposes a task-based macroeconomic model that closes these gaps in a single, numerically solvable switching system. Its ingredients are: a two-layer task model (production and creative tasks) in which the reinstatement rate is driven down endogenously as AGI capability advances, with policy able to offset that decline; a demand side with nominal rigidity, the zero lower bound, and debt-deflation; a fiscal closure in which a capital tax finances workfare (public employment) and transfers under a balanced-budget constraint; and

a social-instability channel that is independent by construction rather than an empirically separated series, a separate equation in which a large purely-relief population generates social pressure. The central methodological challenge is that feasibility here is a question not about a single equilibrium but about whether the high-employment branch still exists: in a piecewise-smooth switching system with a slow fiscal layer and a fast employment layer, no local linearization can answer it, so the transition has to be posed as a bifurcation problem.

Our analytical results are six. We prove that reinstatement collapse drives the labor share monotonically to zero with no self-correcting mechanism, and that skill-retention value is proportional to the reinstatement rate ω, vanishing as $\omega \to 0$ (Theorem 1). We upgrade Fornaro–Wolf's demand shortfall into an explicit, decidable deflationary spiral (Theorem 4), cascading the wage-collapse source into crash dynamics. We establish the existence of a consumption-smooth region and characterize the welfare-determined capital tax τ^*, showing that the Laffer financing ceiling does not bind it even when that ceiling enters the admissible range (Propositions 6–7), and show public employment strictly dominates transfers within a given budget because $1 > \mu_{TM}$ (Proposition 8). We prove that, under the assumed social-pressure law, workfare has a structural protective advantage over a pure unemployment transfer against social-instability risk (Proposition 9), and establish a robustness ceiling for the entire buffer family: beyond a critical shock intensity, no combination of workfare and transfers can prevent an economic crash, so strong shocks require supply-side structural measures (Proposition 10).

Beyond these switching-system results, we develop an analytic bifurcation theory that makes the collapse mechanism itself tractable (Section 4). Deriving the fast subsystem directly from the model, we show that its vector field is quadratic and that the high-employment branch reaches and crosses the switching manifold through a persistence-type boundary equilibrium bifurcation, continuing as a stable low-employment branch on the other side in the frozen fast subsystem; we derive a closed-form critical reinstatement rate and then make that rate endogenous to public employment, workfare and retraining, so that a sufficiently strong policy which retains skill can carry the effective capture across the time-varying threshold, durably only in the no-erosion limit, and use geometric singular perturbation theory to characterize the finite-rate lag of the fast employment variable. The bifurcation is

a static-slice result throughout: on a full path the verified effect of an effective crossing is a discontinuous flattening of the labor-share decline rather than interior convergence, and the distinction is stated explicitly in Section 4.

We anchor three magnitudes to U.S. data (the consumption floor, the transfer MPC $\mu_{TM} = 0.50$, and the workfare wage ratio), set the remaining entries by explicit convention, including the capital-tax growth inhibition $\eta = 0.40$, and run the full battery of experiments under a unified engine (Section 5). On the reported grid the minimum-loss point is a corner solution (workfare-dominant, transfers closed, minimal sufficient capital tax) whose coordinates are grid-dependent rather than identified, and at the baseline frontier growth $g_A = 0.010$ no-policy and pure-transfer regimes are infeasible because they leave the floor unfunded and trigger the modeled MPC-leakage and social channels (Table 1); the consumption-smooth outcome is itself conditional on g_A and is lost once g_A reaches 0.10 (Section 5). No parameter is estimated here, and none of the critical values is presented as one: the reinstatement rate at which the high-employment branch is lost, and the interior workfare share p_E^*, are dimensionless outputs of the calibrated model, so the paper claims structural insight rather than external identification. The central message, read as a model-implied rather than a universal claim, is that a consumption-smooth AGI transition *requires a policy buffer, and that buffer must be financed*.

The remainder of the paper is organized as follows. Section 2 lays out the model. Section 3 derives the analytical propositions. Section 4 develops the analytic boundary equilibrium bifurcation theory. Section 5 presents the calibration and the numerical evidence. Section 6 situates the contributions relative to the literature, and Section 7 concludes.

2. The Model

2.1. Environment

Discrete time $t = 0, 1, \ldots, T$. A continuum of labor $L^S = \bar{L} = 1$ is supplied inelastically and the wage is the numéraire, $w = 1$. Three agents populate the economy: firms that produce with a task technology; a government that taxes capital income and runs transfers / public employment; and households that consume, save, and supply labor. This is a deterministic switching system of aggregate difference equations (neither a DSGE nor an agent-based

model) with no stochastic shocks along the baseline path (Table 3 adds a frontier-growth shock ensemble as a robustness check); all state variables are deterministic time-indexed arrays.

The state variables are the automation threshold I_t, the upper bound of new tasks N_t, the labor-task share $n_t \equiv N_t - I_t$, employment L_t^D, the labor share of employment $s_t \equiv L_t^D/\bar{L}$ (called the labor share throughout; it equals the wage-bill share $w_t L_t^D/Y_t$ only while $w\bar{L} = Y_t$), the price level P_t, nominal debt D_t, real debt $b_t \equiv D_t/P_t$, a skill stock σ_t, and a social-pressure index ς_t.

2.2. Technology: a two-layer task model with an endogenously collapsing reinstatement engine

We extend the Acemoglu–Restrepo task framework to two layers: a production-task layer and a creative-task layer that can itself be automated.

Automation (displacement) frontier.

$$\mathrm{d}I_t^{auto,g} = \kappa_I \, A_t, \qquad A_t = A_0(1+g_A)^t. \tag{1}$$

Reinstatement (new-task creation) engine.

$$\mathrm{d}N_t = \kappa_N^h \, S_t^h + \kappa_N^{AI} \, A_t \, M_0, \qquad S_t^h = S_0 \max\{0,\, 1 - \lambda_h \, I_{t-1}/N_{t-1}\}, \tag{2}$$

where S_t^h is creative labor still performed by humans and κ_N^{AI} governs AGI participation in new-task creation.

Endogenous reinstatement rate. Labor and AGI compete for the new tasks. Let the log capability gain be $g_t = \ln(A_t/A_0)$ and let Γ_t be the policy reattachment stock built by public employment, workfare and retraining, defined in equation (34). The share of new tasks accruing to labor obeys the competitive logistic (discrete-choice) law

$$\omega_t = \Lambda(\ell_0 + \Gamma_t - \beta_A g_t)\,, \qquad \ell_0 = \ln\frac{\omega_0}{1-\omega_0}, \qquad \Lambda(x) = \frac{1}{1+e^{-x}}, \tag{3}$$

where $\beta_A \geq 0$ is the sensitivity of labor's capture to AGI capability. As A_t grows with no policy offset, $\Gamma_t = 0$, the log-odds of labor capture fall and $\omega_t \to 0$: the reinstatement engine collapses endogenously because AGI capability, rather than an assumed low rate, drives the share down. The exogenous treatment is the degenerate case $\omega_t = \omega$, a fixed constant, and

$\beta_A = 0$ gives the policy-only limit in which ω_t rises with Γ_t but does not fall as capability grows.

Task allocation and closed task accounting. Each period's new tasks $\mathrm{d}N_t$ are split between labor and AI by the labor capture efficiency $e_t^{cap} = \omega_t \, \sigma_{t-1}$: labor captures $e_t^{cap} \, \mathrm{d}N_t$ and AI captures the remainder $(1 - e_t^{cap}) \, \mathrm{d}N_t$. The realized direct automation is capped by the labor tasks available, $\mathrm{d}I_t^{auto} = \min\{ \mathrm{d}I_t^{auto,g}, \, n_{t-1} + e_t^{cap} \, \mathrm{d}N_t \}$. The automated stock and labor tasks then obey

$$\mathrm{d}I_t = (1 - e_t^{cap}) \, \mathrm{d}N_t + \mathrm{d}I_t^{auto}, \qquad \mathrm{d}n_t = \mathrm{d}N_t - \mathrm{d}I_t = e_t^{cap} \, \mathrm{d}N_t - \mathrm{d}I_t^{auto}, \tag{4}$$

so the identity $n_t = N_t - I_t$ holds exactly in every period, with $n_t \geq 0$; no independent update of n_t is used. Here $\omega_t \in (0, 1)$ is the *reinstatement rate*, the share of new tasks accruing to labor, determined by equation (3) and reduced to a fixed constant only in the exogenous special case, and σ_{t-1} is the skill stock. **Key modelling choice**: the value of skills is proportional to ω_t. When $\omega_t \to 0$ as AGI takes over new-task creation, so new tasks no longer accrue to labor, skills have no economic return even at arbitrarily high levels. This mechanizes Restrepo's (2025) proposition that skills no longer determine wages.

Accounting scope and leakages. The closure above is exact for the task, population and fiscal flows only, not for a closed national-income account. The construction is partial-equilibrium and supply-side: output GDP_t follows an exogenous supply path, consumption does not feed back into production, so there is no Keynesian demand multiplier, and bank lending capacity is an exogenous credit line rather than a deposit-funded balance sheet. Two flows that the modeled household block does not spend are recorded explicitly as leakages and are not recycled. First, after-tax capital income accrues to capital owners outside the modeled consumption block: on the corner path it cumulates to 265.85, or 81.0% of cumulative GDP, and is treated as capital-owner saving that does not return as demand. Second, the unconsumed share of transfers, $(1 - \mu_{TM})TM_t$, is retained rather than spent: it cumulates to 0.306 on the corner path and likewise does not flow back. A third timing item is the wage accrual. The profit tax base deducts the lagged wage $w_{t-1}L_t^D$ while households receive the current wage $w_t L_t^D$; on the declining path $w_t \leq w_{t-1}$, so the difference $(w_{t-1} - w_t)L_t^D$ is a wage cost accrued for tax but not paid, which cumulates to

0.36 (about 0.11% of cumulative GDP) and stays on the firm side, leaving the tax base conservative rather than overstated. Fully closing these flows would require a capital-owner consumption or investment channel and an explicit recycling rule; absent that channel, treating them as non-returning leakages keeps the flow table complete and is conservative for the demand-sustaining case, since no hidden demand is assumed. The numbers are reproduced by `analysis/check_leakage_accounting.py` and `analysis/check_wage_timing.py`.

Employment and the labor share.

$$L_t^D = \bar{L} \cdot \min\{1,\ \max\{0,\ n_t/N_t\}\}, \qquad s_t = L_t^D/\bar{L}. \tag{5}$$

Skill stock (retraining vs. public employment).

$$\sigma_t = \sigma_{t-1}\Big(1 - \delta_\sigma \tfrac{U_t^{rem}}{\bar{L}} \left(1 - \rho_{ret}\, \beta_{ret}\right)\Big), \qquad \sigma_t \geq \underline{\sigma} = 0.30. \tag{6}$$

Retraining is the only instrument that actively preserves skill, with $\rho_{ret} \in [0,1]$ controlling its scale and $\beta_{ret} \in [0,1]$ how effectively it offsets decay: depreciation in equation (6) is applied only to the unemployed residual U_t^{rem}. Public employment does not actively raise skill and is not a substitute for retraining, but it preserves skill indirectly. A worker on a public job belongs to the employed share rather than to U_t^{rem}, so that worker's skill does not depreciate. Workfare therefore keeps the existing skill stock alive by keeping workers attached to a job, even though it creates no new skill and no market-valued output.

Output.

$$Y_t = (1 + g_{eff})^t \left(1 + g_{disp}\, \delta_{disp}\right), \qquad g_{eff} = g_G(1 - \eta\tau), \qquad \delta_{disp} = (I_t - I_0)/N_0, \tag{7}$$

where g_{eff} embeds the Laffer-style supply-side cost of the capital tax τ.

Capital income (tax base) and taxation.

$$\Pi_t = \max\{0,\ Y_t - w_{t-1} L_t^D\}, \qquad \text{tax}_t = \tau\, \Pi_t. \tag{8}$$

2.3. Demand side: nominal rigidity, the zero lower bound, and debt-deflation

Wage rule (empirical, not a market-clearing condition):

$$w_t = \max\{\underline{w},\ w_{t-1}\,(1 + g_W - \lambda_w\, u_t)\}, \qquad u_t = 1 - s_t. \tag{9}$$

Inflation from the consumption gap.

$$\pi_t = \max\{\pi^{lb},\ \gamma_p\,(C_{t-1}/C^{fe} - 1)\}, \qquad P_t = P_{t-1}(1+\pi_t), \qquad C^{fe} \equiv w\,\bar{L}. \tag{10}$$

Nominal interest and the zero lower bound (ZLB).

$$i_t = \max\{0,\ r_n + \pi_t\}, \qquad r_t^d = i_t - \pi_t. \tag{11}$$

Balance-sheet-consistent household: a joint consumption–debt decision. The household makes one intertemporal choice over consumption and debt rather than choosing consumption first and accumulating residual debt. The disposable real flow in period t is

$$Y_t^d = w_t L_t^D + w^P E_t^P + \mu_{TM} TM_t - \mu_d \max\{0, r_t^d\}, \tag{12}$$

where $w^P E_t^P$ are public-employment wages and $\mu_{TM} TM_t$ are transfers subject to a leakage $\mu_{TM} \in (0,1)$. The stock carried into the period after interest is $D_t^{roll} = D_{t-1}(1+i_t)$. The household then chooses a single net flow, borrowing against an unfunded consumption gap at rate θ and amortizing the debt in excess of the target $\bar{b}$ at rate χ:

$$\Delta D_t^{des} = \theta \max\{0, C^{fe} - Y_t^d\} - \chi \max\{0, D_t^{roll} - \bar{b} P_t\}. \tag{13}$$

Banks supply credit against a hard underwriting ceiling b^{credit} on the debt burden, a genuine stock constraint rather than a consumption cut-off:

$$D_t = \mathrm{clip}\Big(D_t^{roll} + \Delta D_t^{des},\ 0,\ b^{credit} P_t\Big), \qquad \Delta D_t = D_t - D_t^{roll}, \qquad b_t = D_t/P_t. \tag{14}$$

Consumption is the residual of the disposable flow and the realized net borrowing, with no separate truncation:

$$C_t = \max\{0,\ Y_t^d + \Delta D_t/P_t\}. \tag{15}$$

The consumption drag of deleveraging is therefore identical to the real net repayment that actually reduces D, so no demand is suppressed by a term that leaves the stock unchanged. There is no ad hoc cap on the debt burden and no consumption truncation: the only bounds are the bank ceiling b^{credit}, which binds the stock, and the zero floor. Debt-deflation remains active because $b_t = D_t/P_t$ rises when P_t falls, but the endogenous amortization χ and the credit ceiling keep D_t finite, replacing the earlier specification in which a monotone stock with no repayment and no credit supply could diverge. The parameters are $\theta = 0.10$, $\chi = 0.15$, $\bar{b} = 0.35$, and $b^{credit} = 0.80$.

2.4. Fiscal closure: the transition financing gap

Policy response (with delay d):

$$\phi_t = \min\{\phi^{\max},\ \phi^{base} + \rho_{resp}\, u_{t-d}\}. \tag{16}$$

Planned spending.

$$\begin{aligned} E_t^{plan} &= w^P E_t^{P,plan} + \phi_t\, w\, U_t^{rem,plan}, \\ E_t^{P,plan} &= \min\{p_E\, u_t \bar{L},\ u_t \bar{L}\}, \qquad\qquad U_t^{rem,plan} = u_t \bar{L} - E_t^{P,plan}. \end{aligned} \tag{17}$$

Balanced budget / fiscal feasibility:

$$\text{actual spending} = \min\{E_t^{plan},\ \text{tax}_{t-1}\}, \quad \text{shortfall} \Rightarrow \text{proportional scaling } \rho = \text{tax}_{t-1}/E_t^{plan}. \tag{18}$$

The scaling factor $\rho \in (0, 1]$ applies to both components, so the financed public-job headcount and the actual unemployed remainder are

$$E_t^{P,act} = \rho\, E_t^{P,plan}, \qquad U_t^{rem} = u_t \bar{L} - E_t^{P,act}, \qquad L_t^D + E_t^{P,act} + U_t^{rem} = \bar{L}. \tag{19}$$

The remainder is therefore recomputed against the financed headcount rather than the planned one: an unfunded position does not absorb a worker, and the social pressure and skill depreciation equations below use this actual U_t^{rem}, so promised-but-unpaid jobs provide no protection. The population partition closes to machine precision in every period (verified by `analysis/verify_population_accounting.py`).

Financed-headcount residual on the corner path. The unemployed remainder is measured against the financed (actual) public-job headcount rather than the planned headcount. On the exogenous-reinstatement corner path at $t = 5$, the actual remainder is 0.18550, whereas a calculation that ignored budget scaling would give a planned residual of 0.13690; the actual public-job headcount in that period is 0.06342. Both values are produced by the same engine run, and the population partition $L_t^D + E_t^{P,act} + U_t^{rem} = \bar{L}$ closes to machine precision in every period. Appropriations in period t are funded by taxes already collected in $t - 1$: the government budgets against last period's realized revenue, so the fiscal constraint uses tax_{t-1} rather than the current-period tax. This is a within-period rule with no fiscal reserve account: surpluses are not carried forward and deficits are not rolled into debt, so the fiscal

account is separate from the household consumption debt D_t. When revenue is insufficient, spending is scaled down proportionally rather than borrowed. A financing-constraint collapse occurs when the scaled spending is too small to fund the floor, consumption falls below the subsistence level, and the economy crashes.

Naming the transfer. The transfer $\phi_t w U_t^{rem}$ is paid only to the unemployed remainder U_t^{rem}, is conditional on joblessness and bounded by the ceiling $\phi^{\max}$. It is therefore an unemployment transfer rather than a universal basic income: a UBI would be paid unconditionally to the whole population, including the employed, which this model does not implement. We use "unemployment transfer" (and "pure transfer" for the $p_E = 0$ limit) throughout and reserve "UBI" for the external policy literature.

2.5. Social instability (a separate failure channel)

$$\varsigma_t = \alpha_s \, (U_t^{rem}/\bar{L}). \tag{20}$$

The larger the purely-relief population (smaller p_E), the larger the social pressure. If $\varsigma_t > \varsigma^{th}$ for J consecutive periods, a *social-instability collapse* occurs. Public employment converts relief recipients into dignified jobs, reducing pressure. This is the mechanism behind "workfare $>$ pure transfer." The cutoff values $\varsigma^{th} = 0.50$, the persistence requirement $J = 10$, and the coefficient $\alpha_s = 0.60$ are reduced-form model choices rather than externally estimated quantities: ς^{th} has no direct external anchor, while α_s and J are only partially anchored (Section 5.3). The workfare-over-transfer ranking is therefore conditional on these cutoffs. Section 5 reports a systematic sensitivity sweep that quantifies how far each cutoff must move before the corner policy triggers this channel, and Section 5.3 bounds the channel against the empirical ranges.

2.6. Equilibrium and the steady state

Consumption-smooth path. Given parameters and policy $(\tau, p_E, \phi^{\max}, \omega, \dots)$, the six finite-time criteria are: (i) $s_t < s^{th} \equiv 0.05$ in finite time (the employment share falls below the absorbing value); (ii) $\min_t C_t \geq C^{min} \equiv c_0 \, w\bar{L}$; (iii) inflation is not persistently negative; (iv) ς_t does not exceed the threshold persistently; (v) the financing constraint is not violated; (vi)

b_t does not diverge. We call the operational classification a *consumption-smooth transition*, because criterion (iii) is not imposed as an independent veto. The independent gates are (ii) the consumption floor and (iv) the social threshold, while (i), (v) and (vi) are enforced as stated; the persistent-deflation criterion (iii) is measured and reported but nested inside (ii) and (vi).

Operational nesting rule. For every path we report the longest run of sub-target inflation prints, J^{defl}, and the cumulative price-level decline, $P_0 - P_T$. On the endogenous-reinstatement corner (the main model) $J^{defl} = 79$ and the price level falls from 1.00 to 0.205, so the deflation is fully visible rather than hidden by the label. It does not open a separate failure margin. Under the balance-sheet-consistent household of Theorem 4, real debt is bounded by the hard credit ceiling $b_t \leq b^{credit} = 0.80$: once the ceiling binds, the nominal stock $D_t = b^{credit} P_t$ contracts with the price level rather than compounding, and real consumption is held above the floor ($\min_{t \geq 10} C_t = 0.3562 > C^{min}$). The two channels by which deflation could impose a real cost, a Fisher rise in the real debt burden and a fall in real consumption, are precisely the bounded-debt condition (vi) and the floor condition (ii). A path that satisfies (ii) and (vi) therefore carries no additional deflationary welfare loss, and a deflationary spiral is recorded as a collapse only in the Theorem 4 regime where it itself breaks (ii) or (vi), the same event that already fails the operative gates. The numbers are reproduced by `analysis/check_deflation_nesting.py`. Otherwise the outcome is an *economic collapse* (deflationary spiral / debt / consumption below floor), a *social-instability collapse*, or *not-reached*; the labels are ex-post classifications over the full $T = 80$ simulation, not absorbing-state interruptions.

3. Main Results: Analytical Propositions

3.1. Reinstatement collapse drives the labor share monotonically to zero

Theorem 1 (Endogenous Reinstatement Collapse)**.** *Assume (A1)* $\kappa_I > 0$, $A_t = A_0(1 + g_A)^t$, $A_0 > 0$, $g_A > 0$ *(a strictly monotone displacement frontier); (A2) the exogenous human-task component is bounded,* $h_t \leq \overline{h} < \infty$, *and* $\sigma_{t-1} \in [\underline{\sigma}, 1]$*; (A3) the reinstatement rate follows equation* (3)*, so on the no-policy path,* $\Gamma_t = 0$*, it is* $\omega_t = \Lambda(\ell_0 - \beta_A g_t)$ *with* $g_t = \ln(A_t/A_0)$*. If*

for all $t \geq t_0$,

$$\omega_t \, \sigma_{t-1} \, \mathrm{d}N_t < \kappa_I A_t,$$

then n_t *is strictly decreasing, there is a finite* T^* *with* $n_{T^*} = 0$, *and* $s_t = 0$ *for all* $t \geq T^*$.

Proof. (i) The premise holds on the endogenous path. Decompose the new-task flow into its bounded human component and the capability-driven term, $\mathrm{d}N_t = h_t + q_A A_t$, where $h_t = \kappa_N^h S_t^h \leq \overline{h}$ and $q_A = \kappa_N^{AI} M_0$. In the cap-slack region the labor-task drift is

$$\mathrm{d}n_t = \omega_t \sigma_{t-1}(h_t + q_A A_t) - \kappa_I A_t = \omega_t \sigma_{t-1} h_t - \Big(\kappa_I - \omega_t \sigma_{t-1} q_A\Big) A_t.$$

At the baseline calibration $\kappa_I - \omega_0 q_A = 0.0175 > 0$, and $\omega_t \leq \omega_0$ on the no-policy path, so $\kappa_I - \omega_t q_A \geq 0.0175$ in every period. The first term is bounded, $\omega_t \sigma_{t-1} h_t \leq \omega_0 \overline{h}$, whereas the second grows exponentially with A_t. There is therefore a finite t_0 after which the second term dominates and $\mathrm{d}n_t < 0$. (ii) Uniform negative drift. By A1, A_t grows exponentially; by A2, $h_t \leq \overline{h}$ and $\sigma_{t-1} \leq 1$. Set $c_d = \kappa_I - \omega_0 q_A > 0$, which is positive at the baseline calibration. Define

$$t_1 = \max\left\{0, \; \left\lceil \frac{\ln(2\omega_0 \overline{h}/(c_d A_0))}{\ln(1 + g_A)} \right\rceil \right\}.$$

Then $c_d A_{t_1} \geq 2\omega_0 \overline{h}$, and for every $t \geq \max\{t_0, t_1\}$ at which the cap is slack, using the decomposition in (i) and $\omega_t \leq \omega_0$,

$$\mathrm{d}n_t = \omega_t \sigma_{t-1} h_t - \Big(\kappa_I - \omega_t \sigma_{t-1} q_A\Big) A_t \leq \omega_0 \overline{h} - c_d A_{t_1} \leq -\omega_0 \overline{h} < 0.$$

Hence, while the cap is slack, n_t falls by at least $\omega_0 \overline{h} > 0$ per period; starting from $n_{t_1} \leq N_0$, it reaches the cap region in at most $N_0/(\omega_0 \overline{h})$ periods. Once the cap binds, $\mathrm{d}I_t^{auto} = n_{t-1} + e_t^{cap} \, \mathrm{d}N_t$ and $\mathrm{d}n_t = -n_{t-1}$, so n reaches zero in one further step. The interval up to t_1 is finite, hence $T^* < \infty$ and $n_{T^*} = 0$. (iii) Once $n = 0$, $L_t^D = 0$ and $s_t = 0$; the closed identity $n_t = N_t - I_t$ keeps $n_t = 0$ thereafter. □ □

Corollary 2 (Zero-reinstatement limit, $\omega_t \to 0$)**.** *Since* $\omega_t \sigma_{t-1} h_t \leq \omega_t \overline{h} \to 0$ *and* $\kappa_I - \omega_t \sigma_{t-1} q_A \to \kappa_I > 0$, *the premise* $\omega_t \sigma_{t-1}(h_t + q_A A_t) < \kappa_I A_t$ *holds for large* t. *Hence as AGI capability drives* $\omega_t \to 0$, *the labor share vanishes monotonically with* no self-correcting mechanism.

Remark 3. This is consistent with "skills no longer determine wages" (Restrepo, 2025), but here the reinstatement collapse is *endogenous in the rate itself*: ω_t is driven down by the capability term $\beta_A g_t$ in equation (3), rather than fixed at a low constant and left unexplained.

3.2. The deflationary spiral

Theorem 4 (Credit-Rationed Income-Flow Collapse)**.** *Assume (B1) there exist t_0 and $\varepsilon > 0$ such that, for all $t \geq t_0$,*

$$C_{t-1} \leq \Big(1 + \pi^{lb}/\gamma_p - \varepsilon\Big) C^{fe} \approx 0.867\, C^{fe},$$

so that the inflation floor binds (calibrated $\pi^{lb} = -0.02, \gamma_p = 0.15$); (B2) $r_n < -\pi^{lb}$, so the ZLB gives $i_t = 0$ and $P_t = P_{t-1}(1 + \pi^{lb})$ falls strictly; (B3) the bank ceiling binds persistently, $D_t = b^{credit} P_t$, because the desired net flow exceeds available credit; (B4) after the wage bill collapses the disposable flow is bounded by $Y_t^d \leq \bar{Y}$, where labor income has fallen to zero and the residual transfers are limited by tax_{t-1} and the leakage μ_{TM}. Then: (i) $\pi_t = \pi^{lb} < 0$, $i_t = 0$, $r_t^d = -\pi^{lb} > 0$; (ii) the stock does not diverge: $D_t = b^{credit} P_t$ falls with P_t and $b_t = b^{credit}$ stays constant; (iii) under persistent binding the net flow contributed to consumption is

$$\Delta D_t / P_t = b^{credit}\Big(1 - 1/(1 + \pi^{lb})\Big) \equiv -\delta_{lev} < 0,$$

a real deleveraging forced by the shrinking nominal credit line; (iv) $C_t = Y_t^d - \delta_{lev} \leq \bar{Y} - \delta_{lev}$, and when $\bar{Y} - \delta_{lev} < C^{min}$ consumption breaches the floor in finite time, triggering the economic-crash channel.

Proof. (i) $C_t < C^{fe}$ forces the inflation floor (B1); the ZLB gives $i_t = 0$ (B2) and $r_t^d = -\pi^{lb} > 0$. (ii) The binding ceiling gives $D_t = b^{credit} P_t$; since P_t falls and $i_t = 0$, D_t falls while $b_t = D_t/P_t = b^{credit}$ is constant, so the stock is bounded rather than divergent. (iii) With the ceiling binding in both $t-1$ and t, $D_{t-1} = b^{credit} P_{t-1}$ and $D_t^{roll} = D_{t-1}$; hence $\Delta D_t = D_t - D_{t-1} = b^{credit}(P_t - P_{t-1})$ and $\Delta D_t / P_t = b^{credit}(1 - P_{t-1}/P_t) = b^{credit}(1 - 1/(1 + \pi^{lb})) < 0$ because $\pi^{lb} < 0$. (iv) Substituting into the consumption identity gives $C_t = Y_t^d + \Delta D_t / P_t \leq \bar{Y} - \delta_{lev}$; under B4 this lies below C^{min}, so the floor is breached in finite time. The collapse is driven by an insufficient real income flow together with credit rationing, not by an exploding debt stock. □ □

Remark 5. The two conditions cascade: Theorem 1 supplies the source of the gap (wage collapse $wL^D \to 0$), and Theorem 4 characterizes its crash dynamics (ZLB activated, transfers unable to absorb the gap, banks rationing further credit). This upgrades Fornaro–Wolf's "demand shortfall" into an explicit, decidable failure channel in which the crash follows from an insufficient real income flow rather than from an unbounded debt stock. When the floor breach in part (iv) does not occur because the financed transfer holds the real income flow at or above C^{min}, the same mechanics (i)–(iii) still operate but appear only as a falling price level and forced nominal deleveraging, with b_t constant and real consumption intact; that is the consumption-smooth path of Section 2, on which the deflation is nested rather than recorded as an independent failure.

3.3. Existence of a consumption-smooth region and the Laffer interior solution

Proposition 6 (Existence)**.** *If for all t: (C1)* $tax_{t-1} \geq E_t^{plan}$*; (C2)* $C_t \geq C^{min}$*; (C3)* $\varsigma_t \leq \varsigma^{th}$*; (C4)* b_t *bounded, then* $s_t < s^{th} \equiv 0.05$ *in finite time with no crash, which defines a consumption-smooth transition under the nesting rule of Section 2. The implication is a definitional equivalence rather than an existence proof. Two qualifications make its scope precise. First, C1 is the full-funding condition, and no wide-grid point satisfies it strictly in every period: the count of paths with* $tax_{t-1} \geq E_t^{plan}$ *at every t is zero. When C1 fails, spending is scaled down by the within-period rule, and a path is still consumption-smooth when the scaled spending keeps C2–C4 intact, with persistent deflation nested rather than independently vetoed. Second, under that operational criterion the corner* $(\tau = 0.15, p_E = 0.45, \phi^{\max} = 0.02)$ *and 208 of 384 wide-grid combinations are consumption-smooth in direct simulation, so the consumption-smooth region is nonempty as a verified computational fact rather than an analytically assumed one.*

Proposition 7 (Laffer interior solution τ^*)**.** *Once the labor share vanishes,* $\Pi_t \approx Y_t$*. Transition financing capacity*

$$F(\tau) = \tau\,\Pi(Y(\tau)) \approx \tau\,(1 + g_G(1 - \eta\tau))^t(1 + g_{disp}\delta_{disp})$$

is quasi-concave *in* τ*, with a unique peak*

$$\tau^R \approx \frac{1}{t\,g_G\,\eta},$$

decreasing in transition length t and the inhibition coefficient η. The welfare loss $W(\tau) = \sum_t [gap(\tau) + cost(\tau) + \delta_\tau \tau^2 tax(\tau)]$ is strictly convex in τ (gap decreases concavely, deadweight increases convexly), so there is a unique interior optimum $\tau^* \in (0, \tau^R)$.

Proof. Write $G(\tau) = a - b\tau$ with $a = 1 + g_G$, $b = g_G\eta > 0$, and $\Lambda \equiv 1 + g_{disp}\delta_{disp} > 0$ independent of τ, so $F(\tau) = \Lambda\, \tau\, G(\tau)^t$. Differentiating,

$$F'(\tau) = \Lambda\, G(\tau)^{t-1}\Big[a - b(t+1)\tau\Big].$$

The bracketed term is strictly decreasing and vanishes at $\tau_F = a/[b(t+1)] = (1+g_G)/[g_G\eta(t+1)] \approx 1/(tg_G\eta) = \tau^R$. Since $\Lambda G^{t-1} > 0$, $F' > 0$ below τ_F and $F' < 0$ above it, so τ_F is the unique global maximum; moreover $(\log F)'' = -1/\tau^2 - tb^2/G^2 < 0$, hence F is strictly log-concave and quasi-concave. *Numerical note.* With $g_G = 0.03, \eta = 0.40, t = 80$, $\tau_F \approx 1.06 > 1$; on the economically admissible range $\tau \in [0, 1]$ the Laffer ceiling is not binding and F is still increasing, so τ^* is actually pinned by the welfare loss W, not by the financing peak. *Numerical tests of the conditional claims.* Two direct simulations test the two qualifications in the proof. First, the peak enters the admissible range once $\eta \geq 0.424$ at $t = 80$ or once the horizon reaches $t \geq 85$ at $\eta = 0.40$; across $\eta \in \{0.40, 0.50, 0.60\}$ and $t \in \{80, 100\}$, direct minimization of the engine welfare loss places τ^* at 0.10–0.12, always well inside the corresponding peak ($\tau_F \in [0.57, 0.85]$ when it is admissible), so the financing ceiling never binds the optimum even when it lies within the feasible range. Second, removing the decay assumption means minimizing the engine welfare, which contains no such assumption, over the full admissible grid: at the baseline calibration $W(\tau)$ decreases on the left and increases on the right around an interior $\tau^* = 0.12$ with $W(\tau^*) = 62.29$, while W rises to 154.6 at the right edge, so the right-boundary sign is positive and the optimum is interior, not a corner, without imposing decay. The analytic statement that the interior result is conditional absent the decay assumption is therefore retained, and the numerical evidence identifies the interior optimum directly. Both tests are reproduced by `analysis/check_tax_optimality.py`. For welfare, write $W = \sum_t [-B_t + C_t]$ with $B_t' \geq 0$, $B_t'' \leq 0$; we impose the local tax-base freeze used in the convexity argument, treating Π_t as locally constant in τ over a short window, so $C_t = \delta_\tau \tau^3 \Pi_t$ and $C_t'' = 6\delta_\tau \tau \Pi_t \geq 2\delta_\tau \tax_t > 0$. Then $W'' = \sum_t [-B_t'' + C_t''] > 0$, so W is strictly convex. We have $W'(0) = -B'(0) < 0$. The right-boundary sign $W'(\tau^R) > 0$ is

imposed as an explicit decay assumption, $B_t'(\tau) \le c\, G(\tau)^t$ for some $c > 0$, so that beyond the financing peak the marginal benefit decays while the marginal deadweight $C_t' = 3\delta_\tau \tau^2 \Pi_t$ remains non-negligible. Under this decay assumption, the intermediate value theorem gives a unique τ^* with $W'(\tau^*) = 0$, and strict convexity yields $0 < \tau^* < \tau^R$; absent the decay assumption, the interior result is conditional. □

3.4. The p_E^* interior: public employment versus transfers

Proposition 8 (Within-budget superiority of public employment)**.** *Given a tax-budget B split between EP and TM, hold the fiscal amount fixed and compare the marginal contribution to consumption of one fiscal yuan. It is $\mu_{TM} \in (0, 1)$ for transfers (leakage) and 1 for public employment. On this within-budget consumption margin public employment is strictly preferred whenever $1 > \mu_{TM}$; the claim is a marginal, fixed-budget comparison rather than a full welfare ranking, which also nets out the workfare and deadweight costs. The engine welfare ranking is reported in Table 2, including the boundary $\mu_{TM} = 1$ at which the marginal advantage vanishes and the within-budget ordering reverses once the social channel is also disabled. Shifting one unemployed worker from TM to EP:*

$$\Delta C = w^P - \mu_{TM}\phi w > 0, \qquad \Delta(\textit{fiscal}) = w^P - \phi w = w(r - \phi), \qquad \Delta_\varsigma = -\alpha_s/\bar{L} < 0.$$

The direct fiscal change is positive only when $\phi < r = w^P/w = 0.85$; on an equal-outcome basis transfers cost $w^P/\mu_{TM} > w^P$ regardless. The optimal p_E^* equates the marginal gap-compression plus social-risk benefit with the marginal fiscal-plus-welfare cost. Under the explicit end-point assumptions (E0) $MB(0) > MC(0)$ and (E1) $MB(1) < MC(1)$, the loss is U-shaped in p_E and the optimum is interior and finite; otherwise it moves to the corresponding corner.

Proof. One fiscal yuan contributes 1 to consumption under public employment but only $\mu_{TM} < 1$ under transfers, establishing within-budget dominance. With $w^P = rw$, $r = 0.85$, moving one worker from TM to EP gives $\Delta C = w(r - \mu_{TM}\phi) \ge w(0.85 - 0.50) > 0$ and $\Delta_\varsigma = -\alpha_s/\bar{L} < 0$; the direct fiscal change is $w(r - \phi)$, positive only when $\phi < r$, and on an equal-outcome basis transfers cost $w^P/\mu_{TM} > w^P$. For the interior, assume MB and MC are continuous on $[0, 1]$; the net marginal $MB - MC$ is strictly decreasing because $MB' \le 0$

and $MC' \geq 0$. We impose as structural assumptions (E0) $MB(0) > MC(0)$ (at $p_E = 0$ pure relief makes ς maximal and the first workfare unit avoids a high social-risk event) and (E1) $MB(1) < MC(1)$ (at $p_E = 1$ the fiscal bill is maximal and the marginal social benefit of further public employment is zero). Under (E0)–(E1), the intermediate value theorem and strict monotonicity yield a unique $p_E^* \in (0, 1)$ with $MB = MC$, and the loss is U-shaped around it; if either end-point sign fails, the optimum is at the corresponding corner. At the baseline calibration both endpoint signs are confirmed by direct simulation rather than assumed: numerically differencing the welfare loss under a small workfare-share perturbation gives a negative marginal at $p_E = 0$ and a positive marginal at $p_E = 1$, reproduced by `analysis/check_endpoint_signs.py`. □

3.5. Social-instability threshold and the workfare channel

Proposition 9. *With $\varsigma_t = \alpha_s(1 - p_E)u_t$ (when $E_t^P = p_E u_t \bar{L} \leq u_t \bar{L}$), let u^{max} be peak unemployment. Define*

$$\alpha^*(p_E) = \frac{\varsigma^{th}}{(1 - p_E)\, u^{max}}.$$

A sufficient condition for a social-instability crash is $\alpha_s > \alpha^*(p_E)$ and *u_t remains near u^{max} for J consecutive periods, so that $\varsigma_t > \varsigma^{th}$ is sustained as in Section 2.5. A single-period peak below J does not trigger the crash even if $\alpha_s > \alpha^*$. α^* is strictly increasing and convex in p_E, and $\lim_{p_E \to 1} \alpha^* = +\infty$ (universal public employment closes the social channel). Hence raising p_E pushes the social boundary outward.*

3.6. The robustness ceiling of the buffer family

Proposition 10 (Structural ceiling $\bar{\kappa}_I$)**.** *An economic crash occurs iff some t has $C_t < C^{min}$, where*

$$C_t = Y_t^d + \Delta D_t / P_t, \qquad Y_t^d = w_t L_t^D + w^P E_t^P + \mu_{TM} TM_t - \mu_d \max\{0, r_t^d\},$$

and the net-borrowing term is bounded: it is at most θ when the ceiling does not bind, and under a binding ceiling with deflation it equals $-\delta_{lev} < 0$ so that credit cannot add to consumption (Theorem 4). Let κ_I determine the timing $t^*(\kappa_I)$ of wage collapse (Theorem 1: larger κ_I earlier) and let $\tau\Pi_t$ accumulate with $(1 + g_{eff})^t$. Assume the initial tax base is

insufficient to self-finance the floor, $\tau\Pi_0 < C^{min}$*. Define the critical shock*

$$\bar{\kappa}_I = \inf\Big\{\kappa_I : \min_{t\in[0,T]} C_t(\kappa_I) < C^{min} \text{ for all } (p_E, \phi^{\max}, \tau \le \tau^R)\Big\},$$

where $C_t(\kappa_I)$ *is the model consumption under the bounded net-borrowing rule. Then (i) the buffer family protects on* $\kappa_I \in (0, \bar{\kappa}_I]$*; (ii) for* $\kappa_I > \bar{\kappa}_I$ *a crash is unavoidable for* every $(p_E, \phi^{\max})$*; (iii)* $\bar{\kappa}_I$ *is governed by the race between tax-base accumulation and wage collapse,* $A(t^*(\bar{\kappa}_I)) = \mathrm{Req}(t^*(\bar{\kappa}_I))$ *with* $\mathrm{Req}(t^*) = \sum_{t=0}^{t^*} E_t^{plan}$*. Since buffers are finance-constrained, they can only delay a crash on a finite shock interval; strong shocks require supply-side structural measures (lowering* κ_I*, a reinstatement engine, steering technical direction).*

Proof. By Theorem 1, $A_t = A_0(1+g_A)^t$ grows exponentially while the human component $h_t \le \overline{h}$ is bounded and $c_d = \kappa_I - \omega q_A > 0$; the drift decomposition in Theorem 1 then yields a uniform negative drift $-\omega\overline{h}$ after a finite t_1. Hence $t^*(\kappa_I)$ is finite, nonincreasing in κ_I, and $t^* \to 0$ as $\kappa_I \to \infty$. As $t^* \to 0$ the wage term collapses immediately while cumulative financing $A(t) = \sum_{s\le t} \tau\Pi_s$ approaches $\tau\Pi_0 < C^{min}$; since the net-borrowing contribution is bounded above by θ and cannot exceed the rationed credit line, for sufficiently large κ_I we have $\min_t C_t(\kappa_I) < C^{min}$ under every admissible policy. The defining set S is therefore nonempty and $\bar{\kappa}_I = \inf S$ is well defined, with $\bar{\kappa}_I > 0$ by Proposition 6. *Upward-closure lemma.* If $\kappa_I' \ge \kappa_I$ and $\kappa_I \in S$, then $\kappa_I' \in S$. A larger κ_I accelerates displacement without changing the reinstatement engine, so the wage path deteriorates no later and the bounded, rationed net-borrowing term does not improve it; hence $C_t(\kappa_I') \le C_t(\kappa_I)$ after a finite shift, and a crash for every policy at κ_I implies a crash for every policy at κ_I'. Combining the lemma with the definition of infimum gives (i) protection on $\kappa_I \le \bar{\kappa}_I$ and (ii) an unavoidable crash on $\kappa_I > \bar{\kappa}_I$. The ceiling is the race $A(t^*(\bar{\kappa}_I)) = \mathrm{Req}(t^*(\bar{\kappa}_I))$: faster tax-base accumulation raises $\bar{\kappa}_I$ and stronger tax inhibition lowers it. Buffers therefore succeed only on $(0, \bar{\kappa}_I]$; beyond it supply-side structural measures are required. □

Remark 11 (Summary of the analytical results).

Item	Result	Key structure
P1	Reinstatement collapse $\Rightarrow s_t \to 0$, no self-correction	$\mathrm{d}n = \omega\,\mathrm{d}N\sigma - \mathrm{d}I^{auto}$, displacement dominates (Thm. 1, Cor. 2)
P2	Deflationary spiral emerges	ZLB $r_n < -\pi^{lb}$ + gap unabsorbed (Thm. 4)
P3	Smooth region exists; Laffer interior $\tau^* \in (0, \tau^R)$	$F(\tau)$ quasi-concave, W convex (Props. 6, 7)
P4	p_E^* interior (EP$\succ$TM within budget)	$1 > \mu_{TM}$; fiscal/welfare cost rises (Prop. 8)
P5	Under assumed pressure law: social threshold $\alpha^* \uparrow p_E$; $p_E \to 1$ closes channel	$\varsigma = \alpha_s(1 - p_E)u$ (Prop. 9)
P6	Buffer ceiling $\bar{\kappa}_I$; strong shocks need structure	tax-base vs wage-collapse race (Prop. 10)

4. An Analytic Bifurcation Theory: Boundary Equilibrium Bifurcation of the Reinstatement Mechanism

The switching system of Section 2 is solved numerically and its properties are stated as propositions. To make the collapse mechanism itself analytic, we now derive the fast subsystem directly from the model equations rather than postulating a normal form. Two facts follow from that derivation. First, the induced fast vector field is quadratic, not cubic: its third derivative vanishes identically. Second, the high-employment equilibrium does not meet an interior saddle-node but a *boundary equilibrium bifurcation* (BEB), at which it collides with the piecewise-smooth switching manifold created by the truncation of the human-task share and crosses it with stability preserved under the condition stated below. We then close the model by making the reinstatement rate endogenous to public employment, workfare and retraining, which makes the crossing a policy-conditioned outcome rather than an automatic one: it occurs only when the effective capture reaches the time-varying threshold derived below. As before, the reduction is a local statement; the full switching system remains the object that is calibrated, and the numerical illustration is dimensionless rather than an estimated quantity.

4.1. The fast subsystem: a quadratic vector field

Freeze the slowly evolving state, normalize the state variables

$$k_h \equiv \frac{\kappa_N^h}{N}, \qquad q_A \equiv \kappa_N^{AI} M_0, \qquad \mu \equiv \frac{A}{N}, \qquad s \equiv \frac{n}{N}, \qquad \sigma \equiv \text{skill}, \tag{21}$$

and write the growth of the task stock and of the labor tasks as

$$\frac{\dot{N}}{N} = k_h S_0 \Big(1 - \lambda_h (1 - s)\Big) + q_A \mu, \qquad \frac{\dot{n}}{N} = \omega\sigma \frac{\dot{N}}{N} - \kappa_I \mu. \tag{22}$$

This reduction uses the direct automation flow before the availability cap binds, $\mathrm{d}I_t^{auto} = \kappa_I A_t$, which is exactly the high-employment region where the human-task share is active and n is large. Since $\dot{s} = \dot{n}/N - s\,\dot{N}/N$, expansion gives the fast vector field in that region,

$$\boxed{\dot{s} = f_1(s) = -Cs^2 + Bs + A} \tag{23}$$

with coefficients

$$\begin{aligned} C &= k_h S_0 \lambda_h > 0, & \Xi &= k_h S_0 (1 - \lambda_h) < 0, \\ B &= C\omega\sigma - \Xi - q_A \mu, & A &= \omega\sigma(\Xi + q_A \mu) - \kappa_I \mu. \end{aligned} \tag{24}$$

The decisive structural fact is

$$f_1'''(s) \equiv 0, \tag{25}$$

because the leading nonlinearity in (23) is quadratic. A cubic coefficient on which a cusp normal form would rest cannot be induced from these equations. A one-dimensional quadratic field has a fold-type generic unfolding, but here the event is located not by an interior discriminant but by the boundary of the admissible state space, which is precisely why it is a boundary equilibrium bifurcation rather than an interior fold.

4.2. The switching manifold and the piecewise-smooth system

The human-task share is the truncated function

$$S^h(s) = S_0 \max\Big\{0,\, 1 - \lambda_h (1 - s)\Big\}. \tag{26}$$

It is positive only above the switching manifold

$$s_{\text{floor}} = 1 - \frac{1}{\lambda_h}. \tag{27}$$

For $0 \le s < s_{\text{floor}}$ the human share is identically zero, only the AI-driven component remains, and the fast field in this second region is linear:

$$\dot{s} = f_2(s) = \mu\Big(\omega\sigma q_A - \kappa_I - q_A\, s\Big). \tag{28}$$

The two fields join continuously on the manifold, $f_1(s_{\text{floor}}) = f_2(s_{\text{floor}})$, but their slopes differ. Direct subtraction gives the constant jump

$$f_1'(s_{\text{floor}}) - f_2'(s_{\text{floor}}) = k_h S_0\Big(\lambda_h\omega\sigma - \lambda_h + 1\Big). \tag{29}$$

The fast subsystem is therefore a continuous, piecewise-smooth (C^0 but not C^1) system of the standard Filippov type [13].

4.3. Boundary equilibrium bifurcation

Theorem 12 (Boundary equilibrium bifurcation)**.** *For $\mu > 0$ and with σ fixed, a boundary equilibrium bifurcation [54, 53] occurs at*

$$\boxed{\omega_c = \frac{1}{\sigma}\left(\frac{\kappa_I}{q_A} + s_{floor}\right).} \tag{30}$$

The crossing is transverse, $\partial_\omega f_1(s_{floor}) = \mu q_A\sigma > 0$. For $\omega < \omega_c$ the region-I root is virtual (it lies outside the physical region) and the field on the manifold points toward the absorbing boundary $s = 0$; for $\omega > \omega_c$ an interior equilibrium $s^+ > s_{floor}$ exists and is stable whenever the persistence condition $q_A\mu > C\kappa_I/q_A$ derived below holds.

Proof. By continuity, on the manifold

$$f_1(s_{\text{floor}}) = f_2(s_{\text{floor}}) = \mu\Big(\omega\sigma q_A - \kappa_I - q_A\, s_{\text{floor}}\Big). \tag{31}$$

Since $\mu > 0$, the condition $f_1(s_{\text{floor}}) = 0$ is equivalent to $\omega\sigma = \kappa_I/q_A + s_{\text{floor}}$, which gives (30). Differentiating (31) in ω gives $\mu q_A\sigma > 0$, establishing transversality. The unique region-II equilibrium of f_2 is

$$s^{**} = \omega\sigma - \frac{\kappa_I}{q_A}, \tag{32}$$

which equals s_{floor} exactly at $\omega = \omega_c$ and lies below the manifold for $\omega < \omega_c$. The larger root of f_1 likewise equals s_{floor} at ω_c and moves above it for $\omega > \omega_c$. At the boundary equilibrium

the region-II slope is $f_2'(s_{\text{floor}}) = -q_A\mu < 0$, and using (29) with $\omega\sigma = \kappa_I/q_A + s_{\text{floor}}$ the region-I slope is

$$f_1'(s_{\text{floor}}) = -q_A\mu + \frac{C\kappa_I}{q_A}.$$

It is negative exactly under $q_A\mu > C\kappa_I/q_A$. When this persistence condition holds, both one-sided slopes are negative and the equilibrium crosses the manifold with its stability preserved, which is the persistence-type BEB rather than a fold with annihilation. If the condition fails, the region-I branch is repelling and the event is not of persistence type. The calibrated parameters satisfy the condition throughout the reported horizon (Section 4.7). □

The hard lower boundary $s = 0$ contributes a second, structurally simpler event. From (32), the region-II equilibrium enters the physical domain when

$$\omega_0^{(\text{entry})} = \frac{1}{\sigma}\frac{\kappa_I}{q_A}, \tag{33}$$

at which $s^{**} = 0$. Below this value the physical state space contains no equilibrium at all and the trajectory is absorbed at $s = 0$, which is the collapse. The complete static picture is therefore three regimes: absorption for $\omega < \omega_0^{(\text{entry})}$, a stable low-employment region-II equilibrium between the two thresholds, and a stable high-employment region-I equilibrium above ω_c. Figure 1 depicts this three-regime branch structure and the transverse crossing of the field on the manifold.

4.4. Endogenous reinstatement: capability and policy as competing log-odds

The reinstatement rate is the same endogenous variable as in the main model, equation (3), not a separate constant. Public employment, workfare and retraining reattach workers to the production of new tasks, while AGI capability erodes labor's capture. Let EP_t denote public-employment headcount and TM_t workfare and transfer spending; the policy reattachment stock is

$$\Gamma_t = \gamma_{PE}\frac{EP_{t-1}}{L^S} + \gamma_{TM}\frac{TM_{t-1}}{wL^S} + \gamma_R\,\text{retrain}\cdot\text{skill}_{t-1}, \tag{34}$$

and the reinstatement rate is the competitive logistic law

$$\boxed{\omega_t = \Lambda(\ell_0 + \Gamma_t - \beta_A g_t)\,, \quad g_t = \ln(A_t/A_0).} \tag{35}$$

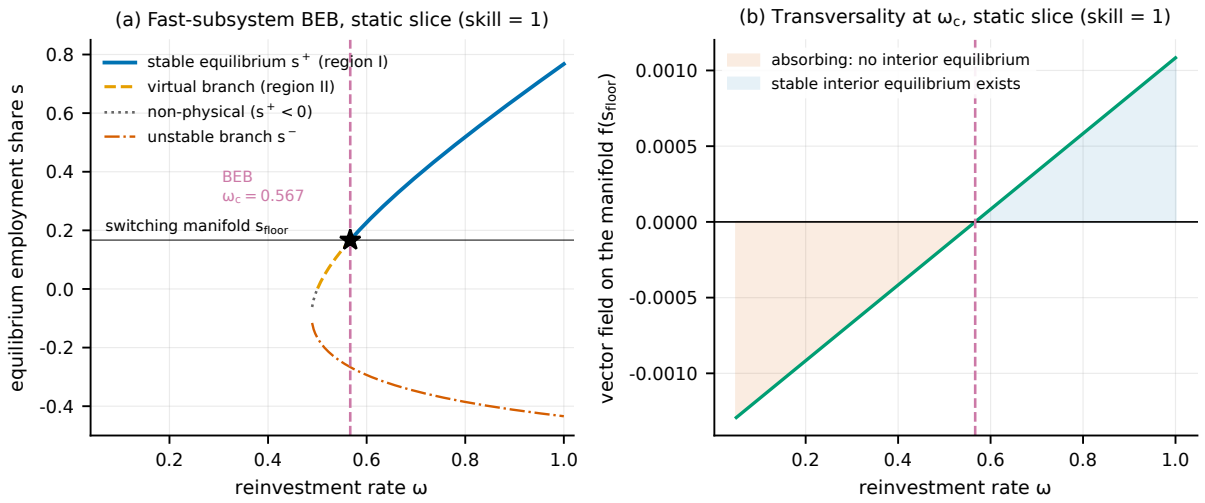


Figure 1: Boundary equilibrium bifurcation of the fast subsystem, static slice with skill fixed at one. Left: the physical equilibrium branch is absorbing at $s = 0$ for $\omega < \kappa_I/q_A$, rises as a stable region-II (low-employment) equilibrium between κ_I/q_A and ω_c, and persists through the switching manifold s_{floor} at $\omega_c = 0.567$ into the stable region-I (high-employment) branch; virtual and non-physical roots are shown light. Right: the field on the manifold $f(s_{\text{floor}})$ crosses zero transversely at ω_c. On a dynamic path skill depreciates, so the effective threshold is the time-varying $\omega_c(t)$ reported in Figure 2.

This specification has the required properties. The logistic keeps $\omega_t \in (0, 1)$ without truncation. With no policy and no capability growth it returns the baseline ω_0; capability growth, $\beta_A g_t > 0$, lowers the rate, and the policy stock Γ_t offsets that decline. Setting $\beta_A = 0$ removes the capability term and recovers the policy-only limit, while the exogenous constant $\omega_t = \omega$ is the degenerate special case used for the calibrated thresholds below.

Proposition 13 (Effective capture and the conditional crossing)**.** *The real bifurcation is governed by the effective capture $\omega_t\, skill_{t-1}$, not by ω_t alone. The instantaneous threshold on the nominal rate is*

$$\omega_c(t) = \frac{\kappa_I/q_A + s_{floor}}{skill_{t-1}},$$

which rises as skill falls. A real crossing into the high-employment region requires the effective capture to reach $\kappa_I/q_A + s_{floor}$, and is therefore conditional on skill being retained. The partial effects of the two competing terms are opposite and bounded, $\partial\omega_t/\partial\Gamma_t = \omega_t(1-\omega_t) > 0$ and $\partial\omega_t/\partial g_t = -\beta_A\omega_t(1-\omega_t) < 0$.

Proof. The fast-subsystem boundary condition is $\omega_t\,\text{skill}_{t-1} = \kappa_I/q_A + s_{\text{floor}}$; dividing by

the lagged skill gives the stated $\omega_c(t)$. Differentiating the logistic in (35) gives the two log-odds derivatives, using the logistic density $\Lambda'(x) = \Lambda(x)(1 - \Lambda(x))$. Since $\text{skill}_{t-1} \leq 1$ on a depreciation path, $\omega_c(t) \geq \omega_c$, so a nominal rate at the static value cannot by itself satisfy the effective condition. □

4.5. Dynamic verification and the critical policy intensity

The conditional crossing of Proposition 13 is verified on the full switching system. Figure 2 contrasts three policy regimes. The capture peaks below are full-path maxima (the no-policy peak occurs at $t = 1$), and each minimum consumption is identical over the full path and the $t \geq 10$ window because every path's minimum occurs after period 10. With no policy, the effective capture never exceeds 0.052, the market employment share is absorbed toward zero and consumption breaks through its floor, with $\min C = 0.057$. Under the workfare corner at $p_E = 0.45$, public employment holds consumption at its floor, $\min C = 0.356$, but the effective capture peaks at 0.121 and never approaches 0.567: the corner defends consumption without restoring the labor engine. Under the strong social floor at $p_E = 0.80$, retraining retains skill, σ ends at 0.96 at $T = 80$, the effective capture crosses the critical value at $t = 35$ under the baseline $\beta_A = 1$, and market employment is retained over this horizon with consumption well above the floor, $\min C = 0.447$; this is a mid-horizon crossing whose long-run durability is qualified by the β_A sweep below. The crossing is therefore conditional on the public-employment intensity: at the weaker standard strength $p_E = 0.45$ the strong-floor instruments raise the effective capture to a peak of 0.498, still below the critical value, so that setting defends consumption without the crossing that the $p_E = 0.80$ configuration achieves.

A bisection on the public-employment intensity gives the sharp threshold at which consumption is defended, defined as $\min_t C_t \geq C_{\min}$:

$$p_E^* = 0.3661, \tag{36}$$

below which the economy crashes and above which consumption is held at its floor. This is a model-implied consumption threshold of the calibrated model, not an identified estimate or a high-employment threshold: at p_E^* the effective capture does not reach the critical value, so consumption is defended by the fiscal buffer without a restoration of market employment. The long-horizon run ($T = 200$) confirms the distinction: with no policy the employment share

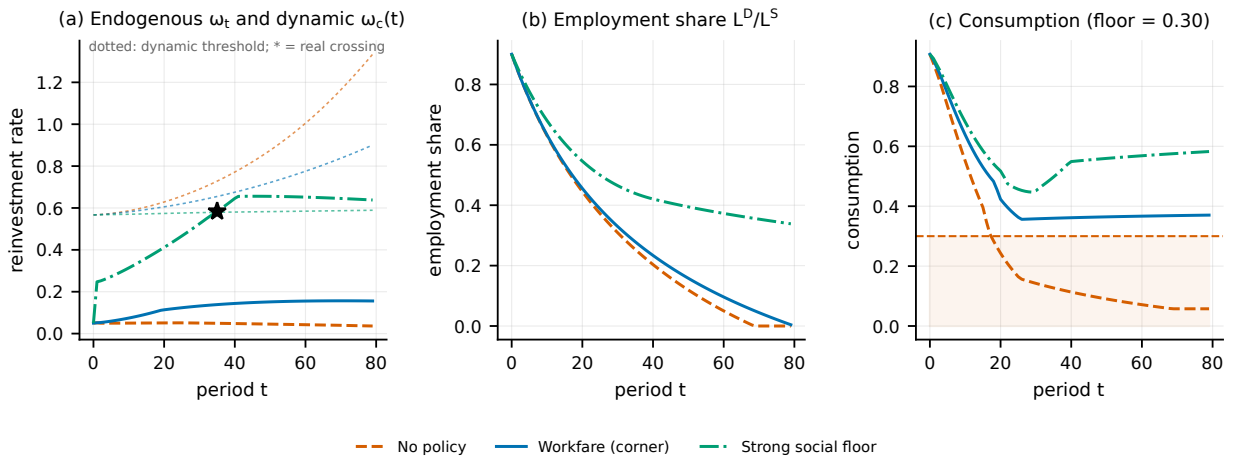


Figure 2: Endogenous reinstatement and the resulting dynamic paths. (a) The endogenous rate ω_t (solid) against its instantaneous dynamic threshold $\omega_c(t)$ (dotted); the strong floor at $p_E = 0.80$ crosses at $t = 35$ under the baseline $\beta_A = 1$ (marked $*$), while the corner and no policy stay below their thresholds. (b) Employment share and (c) consumption; the strong floor retains employment and the highest consumption, the corner holds consumption near the 0.30 floor as market employment vanishes, and the no-policy path collapses.

reaches zero and $\min C = 0.057$, whereas under the corner with transfers closed ($\phi^{\max} \to 0$) the outcome is smooth with $\min C = 0.353$, and under the strong floor at the baseline $\beta_A = 1$ the employment share, after the mid-horizon crossing, is retained at a positive but sub-unity and still-declining share, s_L ends at 0.17, while consumption remains well above the floor with $\min C = 0.447$.

4.6. Slow–fast dynamics and the finite-rate lag

The endogenous rate is a slow variable and the employment share is a fast variable,

$$\dot{s} = f(s, \omega), \qquad \dot{\omega} = \varepsilon\, g(\omega, \text{policy}), \qquad 0 < \varepsilon \ll 1. \tag{37}$$

Away from the switching manifold the critical manifold $f = 0$ is normally hyperbolic, since the fast slope f_s is nonzero, and the standard persistence theory applies [55]. At the boundary equilibrium the two one-sided slopes in (29) are both nonzero but unequal: the field is C^0 rather than C^1, so the normal hyperbolicity is one-sided rather than lost as at a smooth fold, where $f_s = 0$. We therefore do not invoke the fold-type loss of normal hyperbolicity [56, 52, 57]; the piecewise-smooth persistence theory for Filippov systems is the appropriate framework.

The lag seen in the simulations has two sources consistent with this: the slow variable moves at the finite rate ε, so the fast variable tracks a moving equilibrium with a relaxation lag, and crossing the manifold changes the tracking slope discontinuously. It is important to keep the frozen fast subsystem distinct from the full dynamic path. The interior equilibrium of Theorem 12 is an equilibrium of a static slice with skill and the slow variables held fixed; it is not, by itself, an attractor of the full model. On a full path the direct automation flow keeps displacing market labor while the slow variables keep moving, so the verified signature of an effective crossing is a discontinuous flattening of the market-labor-share slope, not a reversal or interior convergence: in the strong regime the mean per-period decline of s_L falls from about -0.009 to about -0.003 across the first effective crossing while remaining strictly negative, and s_L never increases. Interior convergence would require the net automation and reinstatement flows to balance in the long run, a condition the calibrated paths do not reach; even with no capability erosion ($\beta_A = 0$), when the effective capture stays above the threshold durably, s_L keeps declining at a small negative rate and does not settle at a positive level over an 800-period horizon. Reducing ε shrinks the relaxation lag, while raising the level of effective capture moves the static equilibrium itself; the two are distinct instruments, and neither is asserted to produce interior convergence on a full path.

4.7. Connection to the calibrated model

The high-employment region-I equilibrium corresponds structurally to the strong social floor at an intensity sufficient for an effective crossing, while the workfare corner corresponds to the fiscally defended state in which consumption is held at its floor as market employment vanishes, and the absorbing regime corresponds to the no-policy and pure-transfer collapses. The coefficients $\gamma_{PE}, \gamma_{TM}, \gamma_R$, the baseline ω_0 and the capability sensitivity β_A are structural parameters of the reduced subsystem, not externally estimated quantities, and the policy threshold $p_E^* = 0.3661$ is a dimensionless output of the calibrated model under those baseline parameters, illustrative rather than an empirical estimate. The logistic law is the standard share form of a discrete-choice comparison between labor and AGI, so its microfoundation is the competition over new tasks stated in equation (3); a quantitative mapping from the analytic thresholds to an estimated time path remains future work. On the reported corner

path the persistence condition of Theorem 12 holds at every period: the region-I slope $f_1'(s_{\text{floor}})$ stays in $[-0.025, -0.014]$ and is strictly negative, because $q_A\mu$ exceeds $C\kappa_I/q_A$ by an order of magnitude. The persistence-type classification is therefore supported by the calibrated parameters rather than asserted for all $\mu > 0$.

Functional form of the share law. Equation (35) is a modelling choice, so we re-solve the corner with the probit share, $\omega_t = \Phi(\ell_0 + \Gamma_t - \beta_A g_t)$ where Φ is the standard normal cumulative distribution, which shares the logistic form's endpoints, monotonicity and smoothness but a different tail shape. Both forms keep the corner smooth and without a real crossing. Measured over the policy-driven horizon $t \geq 1$, the $t = 0$ print being an initial condition rather than a policy response, the effective capture peaks at 0.121 under the logistic and 0.041 under the probit. The buffer conclusion is nevertheless unchanged, with $\min_t C_t$ moving only from 0.3562 to 0.3571, a gap below 0.001. The binding constraint remains skill depreciation rather than the choice of link function, so the functional form is not the margin that decides the outcome. The capability sensitivity β_A is the more consequential choice and is treated separately as a sensitivity sweep.

Capability-sensitivity sweep. On the $T = 200$ horizon, varying $\beta_A \in \{0, 1, 3\}$ separates a mid-horizon crossing from a durable reinstatement; the peaks quoted are full-path maxima and the critical value is 0.5667. At $\beta_A = 0$, with no capability erosion, the strong floor crosses at $t = 29$ and the effective capture peaks at 0.731 and stays above the threshold, so the crossing is durable in the sense that it is not reversed; the market employment share ends the $T = 200$ horizon at a positive 0.344 without sliding back, although it still declines at a small negative rate rather than settling, as documented over the 800-period horizon in Section 4. At $\beta_A = 1$ the strong floor still crosses, at $t = 35$, with a peak of 0.640 above the critical value, but continued erosion pulls the effective capture back below the threshold later in the horizon, so the employment share keeps declining to 0.172: this is a mid-horizon crossing that defends consumption and employment over the short run without a durable reinstatement. At $\beta_A = 3$ the peak falls to 0.486, below the critical value, there is no crossing at any date and the employment share reaches zero. The corner remains smooth throughout and its minimum consumption stays in 0.356–0.357, while the strong floor's minimum consumption stays in 0.433–0.456 at every reported β_A, so the fiscal buffer defends consumption at every

setting even though a durable restoration of the labor engine occurs only in the no-erosion limit. The baseline $\beta_A = 1$ is the conservative middle setting, not an estimated value.

Counterfactual: the collapse is not built into the logistic term. The law contains a monotone term $-\beta_A g_t$ with $g_t = \log(\text{cap}_t/A_0)$ accumulating over the horizon, so a reader could ask whether the labor-share collapse is imposed by the functional form rather than generated by the model. The direct counterfactual sets $\beta_A = 0$, which removes the erosion channel and leaves only the policy stock and the task accounting, and re-runs the same engine with every other parameter fixed. The no-policy path still crashes, with $\min C = 0.0575$ and the employment share absorbed to zero, and the financed corner still stays smooth with $\min C = 0.356$. The collapse therefore follows from the task-accounting condition $\omega\sigma\, dN < \kappa_I A$ of Theorem 1, which holds even when the reinstatement rate is not being eroded, rather than from the monotone logistic term. The β_A coefficient instead governs the durability of a strong-policy reinstatement, the terminal share 0.344 versus 0.172 reported in the sweep above, not whether the no-policy path crashes. This counterfactual is reproduced by `analysis/check_capability_counterfactual.py`.

5. Calibration

5.1. Parameter–data mapping

Standardization $L^S = \bar{L} = 1$, $w = 1$. In this section the reinstatement rate is held at its exogenous special case, $\omega = 0.05$, so every threshold reported below is the exogenous-reinstatement calibration. The main model endogenizes the rate through equation (3) and reports its own threshold, $p_E^* = 0.3661$; the corner minima there differ slightly from those below, and Section 4.7 reconciles the two. Within the exogenous calibration used here, lowering the transfer ceiling to $\phi^{\max} = 0.001$ at the fixed corner coordinates lowers $\min_t C_t$ from the headline 0.357 at $\phi^{\max} = 0.02$ to 0.353, so the 0.003 gap is the buffer carried by a two-percent transfer ceiling. What the model is anchored to, and where the anchoring stops, is the next matter.

Table conventions. The numerical tables in this section share one reporting standard, stated once rather than repeated. The reinstatement mode is the exogenous special case $\omega = 0.05$ unless a caption states otherwise, while endogenous-mode results use equation (3)

and are flagged in the caption. Every quoted $\min C$ is taken over the evaluation window $t \geq 10$, excluding the initial-condition transient, and the full-path and window minima coincide for the reported paths. Actual fiscal spend is reported in the tables that compare instruments and is the realized, post-scaling spend rather than the planned appropriation. The outcome column uses the three-way classification from Section 2: smooth, crash with its crash type, or not-reached.

What "calibration" means here. Since the term can be read as claiming more than it delivers, we fix its sense before the table. Three entries are anchored to data: the consumption floor ratio c_0 (giving the level $C^{min} = c_0 w \bar{L}$) to the U.S. absolute poverty line, the transfer MPC μ_{TM} to total-consumption MPC estimates for low-liquidity transfer recipients (CARES-era and related transfer studies, not the in-kind SNAP food-at-home margin), and the workfare wage ratio w^P/w to workfare programmes. One group sits inside a literature range and is stated on an annual-frequency convention. The model period is one year, so $\gamma_p = 0.15$ and $\lambda_w = 0.12$ are reduced-form slopes defined directly at the annual rate rather than quarterly slopes carried over without conversion. The quarterly Calvo reference is used only to fix the duration: a quarterly no-reset probability $\zeta_q = 0.75$ maps to an annual no-reset probability $\zeta_y = \zeta_q^4 = 0.316$, a mean annual duration of $1/(1-\zeta_y) = 1.46$ years, while the quarterly duration is four quarters or one year. Nominal rigidity therefore persists for about one year, consistent with adjustment over one or two annual periods; the quarterly structural slope is not mapped period-by-period because these are reduced-form rather than Calvo-NKPC coefficients. The remaining entries are model-set by explicit convention. Among them κ_I carries a distinct label, *calibration target*: it is set so that the $T = 80$ horizon reaches $s_t \to 0$, which is the collapse the paper studies. Matching a parameter to a terminal condition is legitimate practice, but it means that statements of the form "the transition completes within 80 periods" restate the target rather than test it, and κ_I is not an estimated displacement rate. The social channel $(\alpha_s, \varsigma^{th}, J)$ and the auxiliary balance-sheet and accounting coefficients $(\theta, \chi, b^{credit}, \mu_d, \bar{b}, D_0, \delta_\sigma, g_{disp})$ are conventions as well. "Calibrated" therefore means three data-anchored magnitudes plus a two-parameter literature range, with everything else declared as setting; it does not mean that the parameter vector has been estimated, and no result below depends on reading it that way.

Param	Meaning	Baseline	Anchor / source
$s_0 = (N_0 - I_0)/N_0$	initial labor share	0.90	model initial state $I_0 = 0.20$, $N_0 = 2.00$ gives $s_0 = 0.90$; U.S. labor share 53.8% is a benchmark only *[model initial state; see note]*
κ_I	automation displacement	0.020	transition (T=80) drives $s_t \to 0$ *[calibration target; set to hit $s_t \to 0$ at T=80; not estimated]*
g_A	automation frontier growth	0.010	long-run U.S. TFP $\approx 1\%$/yr; the compute-cost decline is not used [34] *[model-set; see note]*
η	capital-tax growth inhibition	0.40	no direct anchor; a growth-inhibition coefficient, not a tax rate [14, 26] *[model-set; see note]*
γ_p, λ_w	nominal rigidity	0.15, 0.12	Smets–Wouters Calvo $\zeta_p = \zeta_w = 0.75$ (Nakamura–Steinsson; Bils–Klenow) [30, 25, 8]; quarterly $\zeta_q = 0.75 \to \zeta_y = \zeta_q^4 = 0.316$ annual, mean duration 1.46 years, and the two slopes are annual-rate reduced-form conventions *[literature range, annual frequency]*
μ_{TM}	transfer MPC (total consumption)	0.50	total-consumption MPC out of transfers: Karger–Rajan EIP 0.46; CARES low-liquidity 0.53–0.66; Cooper–Olivei; IMF 0.20; Miller 0.44–0.55 [11, 12, 18]. The USDA SNAP value 0.30 is the *food-at-home* MPC (the other 0.70 is reallocated across goods, not lost to saving), so it is a different margin and is not used to anchor this total-consumption coefficient [10] *[literature range, total-consumption MPC]*
c_0	consumption floor ratio	0.30	U.S. absolute poverty threshold (ASPE/HHS), divided by the baseline full-time wage income $wL^S = 1$ to give the normalized floor; the historical "food ×3" origin is noted but is not the normalization used here *[data-anchored level; normalization explicit]*
ω	reinstatement rate	0.05	A&R: reinstatement weaker than displacement [2] *[model-set; direction grounded, magnitude not estimated]*
w^P/w	workfare wage ratio	0.85	MGNREGA / World Bank workfare wages set below the market wage (direction grounded);

Note. Three entries of the parameter table carry a "see note" flag. The initial share $s_0 = 0.90$ is computed from the task block itself, $s_0 = (N_0 - I_0)/N_0 = (2.00 - 0.20)/2.00$, so it is an internal consistency requirement rather than a data target; the U.S. labor income share of 53.8% (BLS, 2025Q3, incl. self-employment) serves only as a qualitative long-run benchmark, and the gap reflects that s is a task/employment share rather than a wage-bill share and enters no equation. The frontier growth $g_A = 0.01$ is set at the long-run U.S. TFP rate; the -30%/yr hardware-cost decline and $+40\%$/yr efficiency gain reported by Stanford HAI 2025 / NVIDIA [34] refer to the cost of computing, not to A_t, and are deliberately not used as an anchor (sensitivity in Section 5). The inhibition coefficient $\eta = 0.40$ scales the growth cost of the tax rate in $g_{eff} = g_G(1 - \eta\tau)$ and is not an effective tax rate, so EATR levels (OECD 20.5%, U.S. METR 18.3%) [26] and tax-base elasticities (-0.4 total / -0.7 equipment) [14] serve as magnitudes only. Separately, the symbol σ_t in this paper denotes the skill stock (Section 2.2), not a capital–labor elasticity; the model has no CES production block. The broader elasticity range referenced in task-based calibrations (A&R 2020 P&P ICT $\eta \in (0.5, 0.8)$; K&N > 1; mainstream ≈ 1) is background only and enters no equation [4, 19].

Param	Meaning	Baseline	Anchor / source
α_s	social-pressure coefficient	0.60	$\varsigma_t = \alpha_s U_t^{rem}/\bar{L}$; relief share drives instability *[model-set / calibration convention]*
ς^{th}	social-instability threshold	0.50	sustained $J = 10$-period exceedance $\Rightarrow$ social collapse *[model-set / calibration convention]*
T	simulation horizon	80	transition window over which $s_t \to 0$ is computed *[model-set / calibration convention]*
δ_σ	skill depreciation	0.015	skill stock decay per period unemployed *[model-set / calibration convention]*
g_{disp}	automation output boost	0.05	$\delta_{disp} = (I_t - I_0)/N_0$ output dividend *[model-set / calibration convention]*
θ	borrowing response to gap	0.10	$\theta \max\{0, C^{fe} - Y_t^d\}$ new-borrowing term in the net-flow decision *[model-set / calibration convention]*
χ	amortization of excess debt	0.15	$\chi \max\{0, D_t^{roll} - \bar{b}P_t\}$ net deleveraging rate *[model-set / calibration convention]*
b^{credit}	bank credit ceiling	0.80	hard underwriting bound $D_t \leq b^{credit} P_t$ on the debt burden; a stock constraint, not a consumption cut *[model-set / calibration convention]*
μ_d	monetary drag on disposable flow	0.06	$-\mu_d \max\{0, r_t^d\}$ *[model-set / calibration convention]*
$\bar{b}$	debt burden target	0.35	target real debt in the amortization rule *[model-set / calibration convention]*
D_0	initial nominal debt (normalized)	0.30	initial balance sheet *[model-set / calibration convention]*

Baseline policy levers (headline corner). $\tau = 0.15$, $p_E = 0.45$ (workfare-dominant), $\phi^{\max} = 0.02$ (the grid floor; transfers effectively closed under the baseline loss weights). This is the reported headline, which retains a buffer; the grid's mathematical minimum under the baseline weights sits at $(\tau, p_E) = (0.10, 0.40)$, as explained below.

The transition-loss objective. The grid search ranks policies by a single scalar, and we therefore state that scalar explicitly instead of leaving it implicit in the simulation

code. Writing $C^{fe} = w\bar{L}$ for full-employment consumption, the objective for a policy path $(\tau, p_E, \phi^{\max})$ is

$$\mathcal{L}(\tau, p_E, \phi^{\max}) = \sum_{t=0}^{T} \Big[\max(0, C^{fe} - C_t) + \lambda\, TM_t + \lambda_{PE}\, w^P EP_t + \nu\, EP_t^2 + \delta_\tau\, \tau^2 \mathrm{tax}_t \Big], \quad (38)$$

with $\lambda = 0.5$, $\lambda_{PE} = 0.6$, $\nu = 0.5$ and $\delta_\tau = 0.6$. The four terms are all expressed in consumption units. The first is the consumption gap, the shortfall of realized consumption below what full employment at the prevailing wage would deliver. The second charges transfer outlays TM_t at the leakage rate λ, reflecting that one yuan transferred raises consumption by μ_{TM} rather than by one. The third charges public-employment outlays at the workfare wage, $\lambda_{PE} w^P EP_t$, plus the congestion term νEP_t^2 that makes absorbing a large unemployed pool progressively more expensive. The fourth is the deadweight of the capital tax, quadratic in the rate so that it vanishes as $\tau \to 0$ and produces the Laffer trade-off of Proposition 7. Feasibility is a separate criterion and is not part of $\mathcal{L}$: a path is admissible if $C_t \geq C_{\min} = c_0\, w\bar{L}$ for every $t \geq 10$, the first ten periods being excluded to remove the initial-condition transient, and the statistic $\min_t C_t$ quoted throughout the calibration (including the robustness and social-anchoring subsections) and in Figure 3 refers to that window. Proposition 7 is the one-dimensional projection of (38): with p_E and $\phi^{\max}$ held fixed, the transfer and workfare charges collapse into its single *cost*(τ) term and the deadweight enters with the same quadratic form, so $W(\tau)$ and $\mathcal{L}$ rank policies identically along the τ axis. The four weights are conventions rather than estimates: they set the relative price of a consumption shortfall, a fiscal outlay and a tax distortion, and no external data pin them down. What they do and do not control is reported below. Accordingly, $\mathcal{L}$ is a scenario loss index rather than a normative welfare measure: it ranks parameter-and-rule scenarios under the stated weights, and its values are not a social-welfare level or a compensating or equivalent variation, because no household utility function is specified. The normative statements are therefore confined to the feasibility and financing conditions, while the loss numbers are comparative scenario statistics.

External ranges for the two policy instruments. Both instruments enter the closure mechanically, and two reduced-form ranges in the empirical literature can check those settings without becoming inputs. For workfare the relevant object is the employment multiplier: Lee, Ko and Kim [59] exploit public-sector relocations in South Korea as an exogenous source of public

employment and estimate that one public-sector position raises private-sector employment by one, with the effect highly localized. Gechert [58] reports a meta-regression over 104 studies in which public-spending multipliers are close to one and exceed those of tax cuts and transfers by 0.3–0.4. Our closure contains the direct positions and the demand they generate through the consumption block, but not a localized private-employment multiplier; the published one-for-one estimate therefore sits above what we model, and adding it would raise rather than lower the workfare ranking of Proposition 8. For the pure transfer the relevant object is the labor-supply response. Vivalt et al. [60] report, from the OpenResearch experiment that paid $1,000 per month for three years to 1,000 low-income adults against 2,000 controls, a 2.0 percentage-point fall in labor-market participation and a 1.3–1.4 hour reduction in weekly hours. Our model has no hours margin, so a transfer leaves labor supply unchanged; that omission flatters the pure transfer, since a smaller effective labor supply would shrink the tax base that has to carry the buffer. Both gaps therefore run in the direction that makes the ranking harder, not easier, to establish: the missing multiplier weakens workfare and the missing labor-supply response strengthens the pure transfer. Filling either with the published range moves the comparison the same way. Neither range enters an equation, and this is the sense in which the policy claims are model-implied rather than universal.

5.2. Robustness of the analytical results under the calibrated parameters

Figure 3 reports the baseline path under the corner configuration. Under the calibrated parameters ($\mu_{TM} = 0.50$, $\eta = 0.40$), transfers are costly relative to public-employment wages (one yuan of transfers yields 0.50 of consumption, whereas a public-employment wage yields 1), and the capital tax carries a Laffer supply-side cost. Running the full battery under the unified model engine yields the following calibrated outcomes:

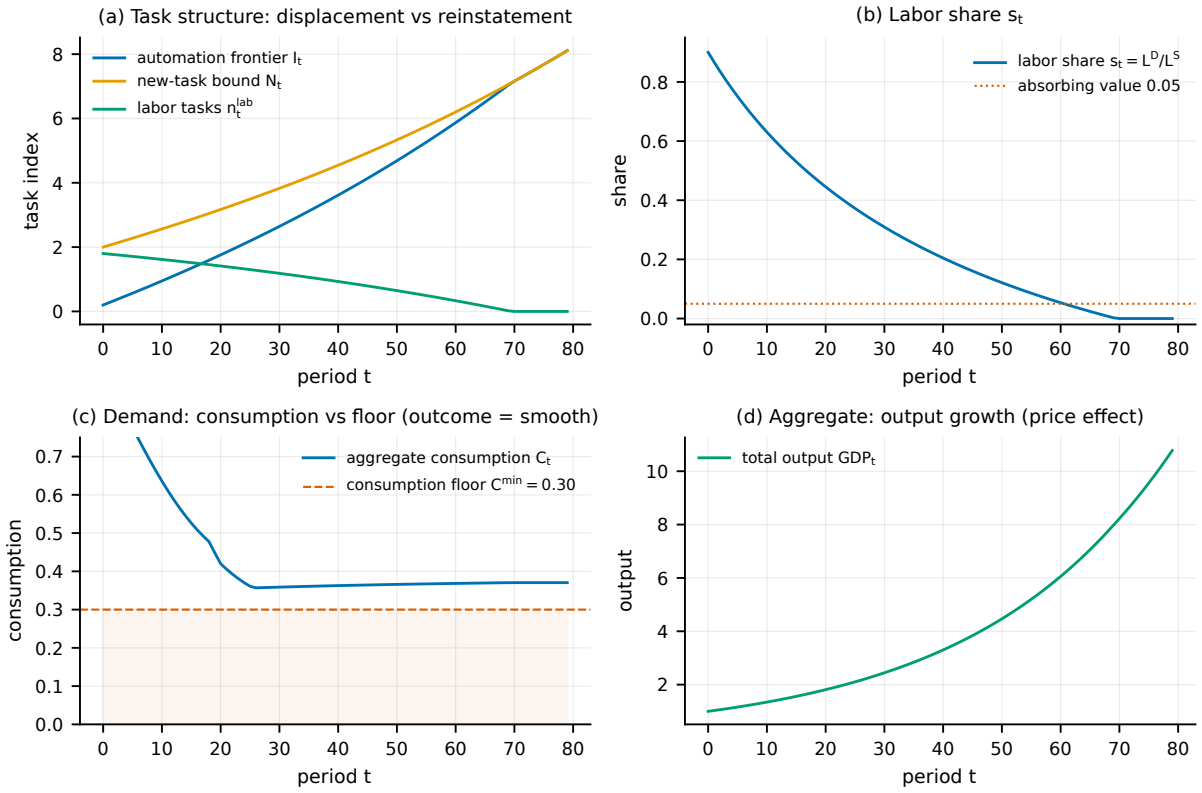


Figure 3: Baseline smooth transition under the corner configuration ($\tau = 0.15, p_E = 0.45, \phi^{\max} = 0.02$) at the baseline frontier growth $g_A = 0.010$. (a) The automation frontier I_t and new-task bound N_t diverge while labor tasks n_t^{lab} vanish. (b) The labor share declines monotonically to its absorbing value. (c) Aggregate consumption stays above the floor $C^{min} = 0.30$ (minimum 0.357). (d) Total output keeps rising through the transition.

Result	Calibrated outcome	Conclusion
Policy delay d	corner ($\phi^{\max}$=0.02): d=0 and d=12 both smooth (0.357); standard ($\phi^{\max}$=0.65): d=0 and d=12 both smooth (0.358 and 0.366)	delay immaterial at the corner and under the standard setting
Isolation gain Δ vs ω	$0.018 \to 0.514$	monotone, robust
Workfare vs pure transfer	6/6 not crashed (4 smooth/reached, 2 not-reached) vs 6/6 crashed	workfare > transfer at every ω
Minimum-loss point (grid)	$(0.10, 0.40, 0.02)$	corner: workfare-dominant, close transfers, low tax
Feasible share, p_E=0.45 vs 0.30	64.1% vs 26.6%	advantage retained
No-policy / pure-transfer	crash / all crash	buffering and financing necessary

The feasible share is the fraction of the 8×8 $(\tau, \phi^{\max})$ cells at the stated p_E that are smooth under the operational criterion in Proposition 6, evaluated on the same wide grid: $41/64 = 64.1\%$ at $p_E = 0.45$, $17/64 = 26.6\%$ at $p_E = 0.30$, and $208/384 = 54.2\%$ over the grid as a whole. The workfare–transfer row is a same-fiscal comparison: the two regimes use the identical ($\tau = 0.16, \phi^{\max} = 0.65$) and differ only in p_E (0.45 public employment versus 0 pure transfer). Across six ω values, workfare survives all six ($\omega = 0.02$ and 0.05 hold the floor at 0.351 and 0.358), the pure transfer crashes in all six, and workfare carries the lower total loss in all six. Of the six surviving workfare paths, four reach the employment target ($\omega = 0.02$–0.20) and two are not-reached ($\omega = 0.30$ and 0.50), so the no-crash count is not a claim that all six reach the target.

The public-employment intensity is load-bearing. The same six ω values at the corner $(0.15, 0.02)$ survive at $p_E = 0.45$ ($\min_t C_t = 0.352$–0.357) but all collapse at $p_E = 0.30$ ($\min_t C_t = 0.244$–0.265, every path through the floor). The corner survives only above a public-employment share of about 0.36, a feasibility bound rather than a tuning preference. The decomposition separates the two mechanisms rather than attributing the result to financing alone. With an equal consumption MPC the transfer holds consumption above the floor ($\min C = 0.4086$), so consumption support under a pure transfer is feasible once

Table 1: Structural decomposition of the pure-transfer failure. All four runs use the identical fiscal setting ($\tau = 0.16, p_E = 0, \phi^{\max} = 0.65$) and vary only the two structural choices the workfare corner does not share: the transfer consumption MPC μ_{TM} and the social-instability coefficient α_s. Reproduced by `analysis/check_transfer_structure.py`.

Configuration	outcome	crash type	$\min C$
Baseline ($\mu_{TM} = 0.50, \alpha_s = 0.60$)	crash	both	0.2589
Equal MPC ($\mu_{TM} = 1.00$)	crash	social only	0.4086
Social disabled ($\alpha_s = 0$)	crash	economic	0.2589
Both relaxed ($\mu_{TM} = 1.00, \alpha_s = 0$)	smooth	none	0.4086

the MPC leakage is removed, and the remaining failure is the social channel. With the social channel disabled but the MPC leakage retained, the path still crashes economically ($\min C = 0.2589$), because only half of each transfer yuan reaches consumption. Only when both structural choices are relaxed does the regime become smooth. The workfare advantage therefore rests on the combination of the unit workfare MPC and the social relief-to-jobs conversion, not on a generic financing failure: the table states the exact conditions under which the ranking holds, and the replication script reproduces all four rows.

Table 2: Symmetric-limit welfare comparison on the same fiscal setting ($\tau = 0.16, \phi^{\max} = 0.65$), with the transfer MPC set to one and the social channel disabled ($\mu_{TM} = 1, \alpha_s = 0$). The two columns differ only in the public-employment share p_E. Total loss is the engine welfare loss and fiscal spend is the transfer plus workfare wage bill. Reproduced by `analysis/check_symmetric_welfare.py`.

Regime	outcome	total loss	fiscal spend	$\min C$
Public employment ($p_E = 0.45$)	smooth	55.34	36.12	0.4177
Pure transfer ($p_E = 0$)	smooth	51.23	32.01	0.4086

The symmetric limit reverses not only the feasibility classification but also the welfare ranking. On the identical fiscal setting with $\mu_{TM} = 1$ and $\alpha_s = 0$, both regimes are smooth, yet the pure transfer attains a lower total loss (51.23 against 55.34) at a lower fiscal spend (32.01 against 36.12). The mechanism is the removal of the two margins on which workfare dominance is built. Once the transfer MPC equals one, a transfer yuan supports consumption

exactly as a workfare wage does, and once the social channel is disabled, the relief-to-jobs conversion no longer avoids a social-risk event; the public-employment path still carries the workfare cost, including its quadratic term, while the transfer does not. The consumption-gap component is slightly smaller under public employment, but its higher policy cost dominates, so the net loss is higher. This is a conditional result rather than a contradiction of the baseline ranking: the reversal requires both $\mu_{TM} = 1$ and $\alpha_s = 0$ simultaneously. The calibrated values $\mu_{TM} = 0.50$, anchored to measured transfer spending responses, and $\alpha_s = 0.60$, a deliberately conservative setting, satisfy neither, and under them the workfare corner keeps both the lower loss and the feasible outcome reported in Table 1. The symmetric numbers should therefore be read as the boundary at which the policy ranking changes, not as the baseline case.

Reading this section. Four of the analytical results are visible directly in the runs below: reinstatement collapse drives $s_t \to 0$, skill-retention value decays monotonically in ω (Figure 6), workfare ranks above the pure transfer in the modeled crash channels, and the protective combination widens the surviving region. Figure 4 contrasts the three regimes on the consumption margin, and Table 1 documents why the no-policy and pure-transfer regimes leave the floor unfunded.

Sensitivity to g_A and η. Two calibration entries admit no verifiable external anchor, the automation-frontier growth g_A and the capital-tax growth inhibition η, so we sweep both and report the outcome instead of asserting that the baseline values are innocuous. Varying g_A over $\{0.010, 0.05, 0.10, 0.20, 0.30\}$ at the corner gives minimum consumption $0.357, 0.344, 0.294, 0.237, 0.222$ and outcome smooth, smooth, crash, crash, crash. The financed buffer therefore holds at $g_A = 0.05$, one sixth of the -30%/yr hardware-cost decline often quoted for computing; the floor is breached only at $g_A = 0.10$, and at $g_A \geq 0.20$ the transition crashes hard with a rapidly accumulating loss. This remains the sharpest sensitivity in the calibration, and the smooth conclusion should be read as conditional on a conservative long-run-TFP frontier growth rate rather than on a hardware-cost rate. By contrast η is nearly inert over the admissible tax range: sweeping η from 0.20 to 0.80 leaves every corner outcome smooth at $\min_t C_t = 0.357$ and moves the total loss by about 0.1%, so the imprecision of η affects no conclusion, while also implying that no sharp Laffer effect can

be claimed at $\tau = 0.15$. Both sweeps are reproducible from the replication script with the corner configuration.

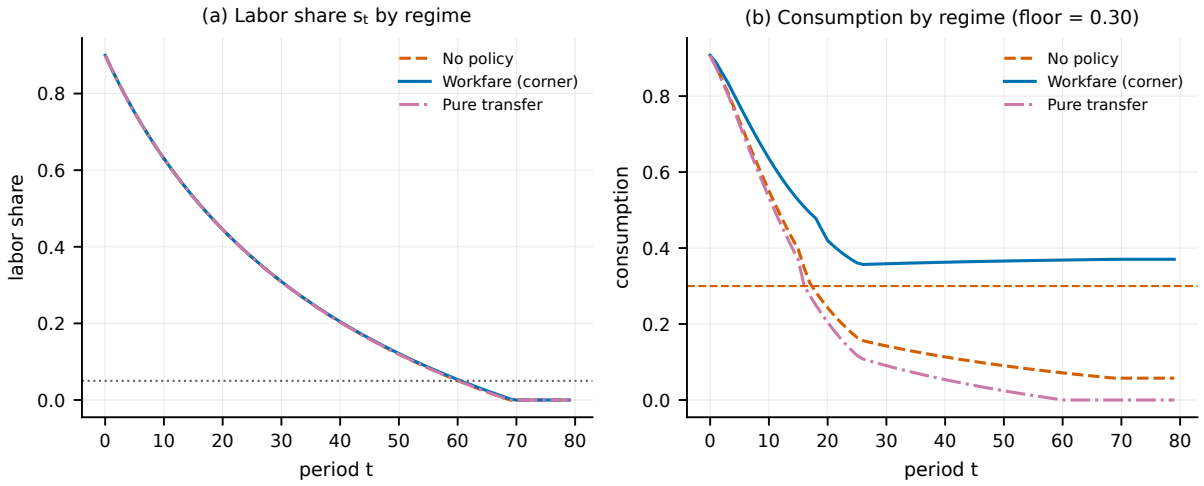


Figure 4: Policy-regime comparison. Each regime carries its own fiscal setting: no policy ($\tau = 0.05, \phi^{\max} = 0.15$), the workfare corner ($\tau = 0.15, p_E = 0.45, \phi^{\max} = 0.02$), and the pure transfer ($\tau = 0, \phi^{\max} = 0.65$). This is therefore a comparison of regimes rather than a same-fiscal instrument test, which is given in the table above. (a) The labor share s_t follows the same task-driven decline under all three regimes. (b) The workfare corner holds aggregate consumption above the floor, whereas the no-policy and pure-transfer regimes fall below it and become infeasible; the consumption buffer, not the market employment count, is what distinguishes the regimes.

Grid minima and identification. Figure 5 visualizes the loss surface. Three minima are reported. A scan that fixed $\phi^{\max} = 0.15$ returns $(0.18, 0.40, 0.15)$ with loss 63.53; the wide grid, which extends $\phi^{\max}$ down to 0.02, has its mathematical minimum at $(0.10, 0.40, 0.02)$ with loss 62.84; and a denser grid that extends $\phi^{\max}$ to 0.001 returns $(0.10, 0.40, 0.001)$ with loss 62.82, taken up below. The three are ordered monotonically in the ceiling (the lower the cap on transfers, the lower the minimized loss), with a total spread of about 1.1%. The loss surface is locally flat, so the exact coordinates are grid-dependent rather than sharply identified. The wide-grid minimum $(0.10, 0.40, 0.02)$ clears the floor by only four thousandths ($\min_t C_t = 0.304$ against a floor of 0.300), so we report the better-buffered corner $(0.15, 0.45, 0.02)$ with loss 63.14 and $\min_t C_t = 0.357$ as the headline; this is a deliberately conservative choice rather than the grid's mathematical minimum, and the solid star in Figure 5 marks the actual minimum $(0.10, 0.40)$.

A refinement test addresses the boundary directly. Re-solving on a denser grid (capital-tax step 0.01, workfare step 0.02, and $\phi^{\max}$ extended down to 0.001) moves the minimizer to $(0.10, 0.40, 0.001)$ with loss 62.82. Against the buffered headline corner 63.14 this is a relative difference of 0.51%, whereas the wide grid's mathematical minimum is 62.84 at the same $(0.10, 0.40)$ coordinates, so the extra refinement changes the loss only in the third decimal once the headline buffer is set aside. The transfer ceiling is monotone in the same direction: holding $(\tau, p_E) = (0.15, 0.45)$, lowering $\phi^{\max}$ from 0.10 to 0.001 lowers the loss from 63.19 to 63.13 and $\min_t C_t$ from 0.371 to 0.353. Closing transfers is therefore a preference of the objective rather than a grid-resolution artifact, but two caveats remain. The exact corner coordinates are not identified, and the denser solution carries a thin buffer: $\min_t C_t = 0.3070$ against a floor of 0.300, so it clears by about seven thousandths. The refined solution is reported here to show what the objective does once the grid stops being the binding constraint.

Sensitivity to the loss weights. The weights in (38) are conventions, so we report what they do and do not control rather than asserting that they are innocuous. Feasibility is weight-independent by construction: re-ranking all 384 grid points under six weight configurations leaves the admissible set at 208 points in every case, so whether a path is smooth or crashes does not depend on λ, λ_{PE}, ν or δ_τ, and the feasible shares reported above are unaffected. The ranking among admissible paths does depend on them. The configuration $(0.10, 0.40, 0.02)$ is selected in three of the six configurations (the baseline, flat and tax-heavy weights). The remaining configurations move the minimizer: halving the cost weights relative to the gap gives $(0.20, 0.55, 0.65)$ with $\min_t C_t = 0.431$, and doubling them gives $(0.12, 0.30, 0.20)$ with $\min_t C_t = 0.303$. In every configuration public employment remains the dominant instrument, with $p_E \in [0.30, 0.55]$, whereas the transfer ceiling ranges from 0.02 to 0.65. Workfare dominance is thus a property of the model, while the statement that transfers are closed at the optimum is a property of the baseline weights together with the lower grid bound, and is reported as such.

Stochastic robustness. The analytical results are deterministic, and a deterministic path should not be presented as if it stood in for a stochastic one. We therefore shock the frontier growth itself, the parameter on which the smooth conclusion most depends. The growth rate becomes $g_{A,t} = 0.010 + \varepsilon_t$ with an AR(1) deviation ($\rho = 0.5$, Gaussian innovations

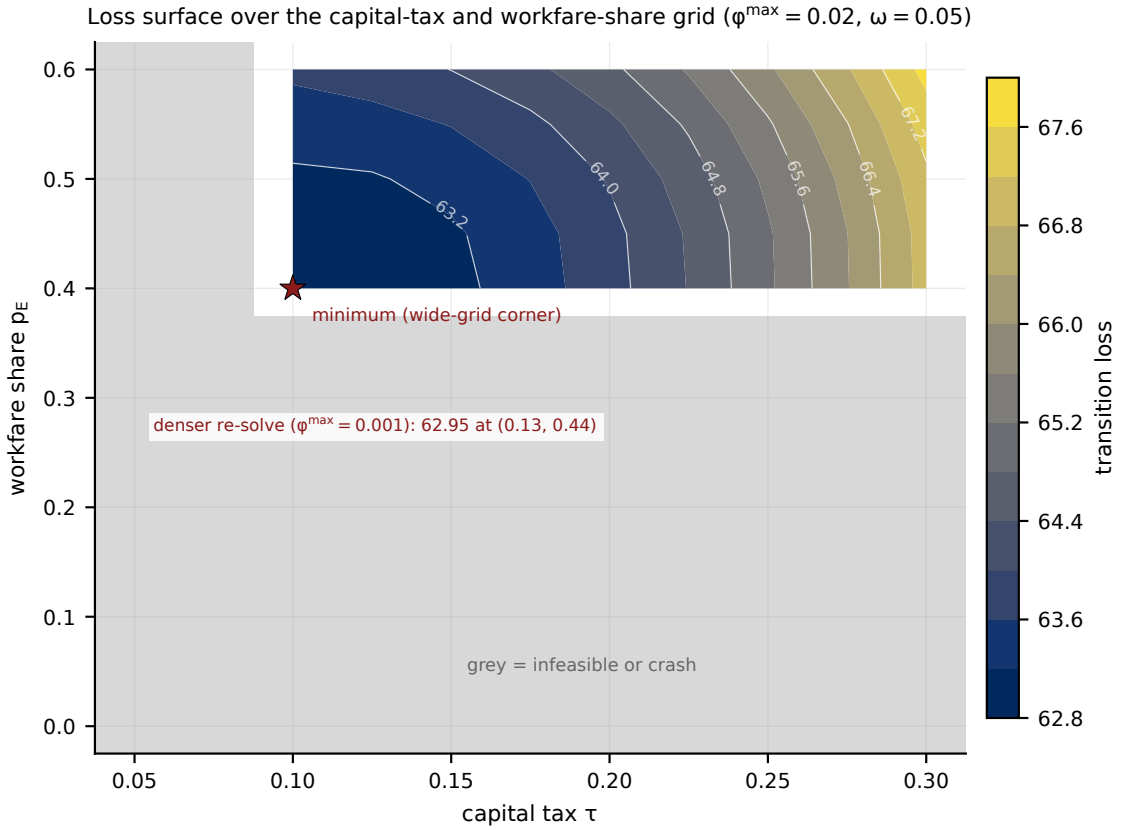


Figure 5: Transition loss over the capital-tax and workfare-share grid at $\phi^{\max} = 0.02$, $\omega = 0.05$. The grey region is infeasible (crash); the smooth region is confined to high workfare shares, and under the baseline loss weights the loss-minimizing point on this grid sits at its corner (solid star). The smooth surface is locally flat, so the minimizer's exact coordinates are grid-dependent rather than identified. A denser re-solve that extends $\phi^{\max}$ to 0.001 moves the minimizer to $(0.10, 0.40)$ with a loss of 62.82 (stated in the panel); that location is not a point on the $\phi^{\max} = 0.02$ surface drawn here, which is why it is reported in words rather than marked.

of standard deviation σ_ε (the shock scale, distinct from the skill index σ_t)), the engine is otherwise unchanged, and each configuration is run 200 times; a path is recorded as a crash when it triggers the economic or social crash test before the horizon ends. Table 3 reports the results.

Table 3: Monte Carlo crash counts under frontier-growth shocks (200 replications; AR(1), $\rho = 0.5$). Buffer is $\min_t C_t - 0.300$; the last column reports the fifth percentile of $\min_t C_t$ at $\sigma_\varepsilon = 0.03$.

Configuration	buffer	σ_ε=0.01	σ_ε=0.02	σ_ε=0.03	$\min_t C_t$, 5%
Headline $(0.15, 0.45, 0.02)$	0.056	0/200	0/200	0/200	0.3540
Refined $(0.13, 0.44, 0.001)$	0.048	0/200	0/200	0/200	0.3348

Two things follow. First, neither configuration produces an economic crash across the shock sizes considered, because the balance-sheet-consistent household block keeps the debt stock bounded and the financed buffer defends consumption. With zero crashes in 200 draws, the Clopper–Pearson one-sided 95% upper bound on the true crash probability is 1.49%, so zero observed crashes does not establish a zero crash probability. At $\sigma_\varepsilon = 0.03$, however, 69 of the 200 headline paths (65 of the refined paths) fail to drive the labor share to its absorbing value within T because the shock raises the frontier; these are right-censored non-absorption outcomes, counted as neither survival nor crash, and they are not evidence of robustness, since the horizon ends before either state is reached. Second, the margin still separates the two configurations even though neither crashes. The headline corner, with a buffer of 0.056, holds its fifth-percentile minimum consumption at 0.354; the refined minimum-loss configuration $(0.13, 0.44, 0.001)$, which clears the floor by 0.048, sees its fifth-percentile minimum consumption fall to 0.335. The refined cushion is therefore thinner and leaves consumption closer to the floor under shocks, even though the bounded debt block prevents a hard crash at the scales tested.

The deterministic v6 global-best point $(0.10, 0.40, 0.001)$ is a different case and should not be read as a third crash-free stochastic design. Its deterministic buffer is only 0.007, and when the same point is run through the stochastic v12 engine it does crash: 16 paths at $\sigma_\varepsilon = 0.01$, 58 at 0.02 and 74 at 0.03, with fifth-percentile minimum consumption falling to 0.298, 0.288 and 0.279. The mechanism is that a seven-thousandth buffer cannot absorb the

cumulative AR(1) deviation of the frontier; on the crashing paths consumption first breaks the floor around the middle of the horizon after the skill stock has already depreciated. This separates the two refinements cleanly: the v12 refined point $(0.13, 0.44)$ keeps a 0.048 buffer and stays crash-free, whereas the v6 point $(0.10, 0.40)$ trades that buffer for a marginally lower deterministic loss and fails once shocks are present. Buffer thickness, rather than the third decimal of the deterministic loss, decides stochastic survival. This is the quantitative counterpart of the qualitative statement above: the buffer buys time, and its thickness determines how much. Demand-side, monetary and policy-implementation shocks are not modelled, and a fully stochastic version of the model remains for future work (Section 7).

Two configurations, not one. The smooth headline depends on the public-employment intensity rather than on the transfer ceiling. At the baseline $\omega = 0.05$, the wide-grid corner $(0.15, 0.45, \phi^{\max} = 0.02)$ runs smooth with $\min C = 0.357 > C^{min} = 0.30$. Under the standard setting $(\tau = 0.16, p_E = 0.45, \phi^{\max} = 0.65)$ used in the social-channel comparison of Table 4, the point $(0.16, 0.30, 0.65)$ runs smooth at $\min C = 0.333$, while $(0.16, 0.45, 0.65)$ runs smooth at $\min C = 0.358$. The binding condition is therefore the workfare share p_E, not a tight transfer ceiling: under the standard setting both workfare shares clear the floor, and a higher share raises the consumption margin. The sharp threshold separating smooth from crash for this standard calibration is $p_E = 0.167$, below the 0.30 point.

Social-threshold sensitivity sweep. A dedicated social-channel sweep uses the model engine and varies one social-channel parameter at a time: ς^{th} over 0.20–0.70, α_s over 0.30–1.20, and J over 3–30, plus a two-dimensional $\varsigma^{th} \times \alpha_s$ grid, under the calibrated parameters $(\eta = 0.40, \mu_{TM} = 0.50)$. Three policies are compared: the recommended corner $(\tau = 0.15, p_E = 0.45, \phi^{\max} = 0.02)$, the standard setting $(\tau = 0.16, p_E = 0.45, \phi^{\max} = 0.65)$, and the pure transfer $(p_E = 0)$ as the extreme social-channel case. Table 4 reports baseline outcomes; Table 5 reports the stability/crash boundaries.

The two-dimensional grid for the corner (S = social crash, . = no social crash) traces the same boundary:



Figure 6: Isolation experiment: the loss saved by retraining Δ rises monotonically with the reinstatement rate ω, the share of new tasks accruing to labor. As $\omega \to 0$ (AGI takes over new-task creation and reinstatement fails), the value of retraining vanishes, confirming that skill retention pays off only while labor can still capture new tasks.

Table 4: Baseline social-channel outcomes ($\varsigma^{th} = 0.50, \alpha_s = 0.60, J = 10$, $g_A = 0.010$). Values within 10^{-3} of the floor are given to four decimals.

Policy	outcome	crash type	min C	peak stress	social crash
Corner $(0.15, 0.45, 0.02)$	smooth	none	0.357	0.330	no
Standard $(0.16, 0.45, 0.65)$	smooth	none	0.3576	0.330	no
Pure transfer $(0.16, 0, 0.65)$	crash	both	0.2589	0.600	yes

$\varsigma^{th} \backslash \alpha_s$	0.45	0.60	0.75	0.90	1.20
0.25	.	S	S	S	S
0.30	.	S	S	S	S
0.33	.	.	S	S	S
0.40	.	.	S	S	S
0.50	.	.	.	.	S

Mechanism. At $p_E = 0.45$, public employment absorbs 45% of the unemployed and only 55% remain on pure relief, so $\varsigma_t = \alpha_s(1 - p_E)u_t = 0.55\,\alpha_s u_t$, peaking at $0.55\,\alpha_s\,u^{max} = 0.330$

Table 5: Social-crash boundaries under one-at-a-time sweeps.

Swept parameter (baseline)	Corner	Standard	Pure transfer
critical ς^{th} (base 0.50)	0.33	0.33	0.60
critical α_s (base 0.60)	0.91	0.91	0.50
critical J (base 10)	never triggers	never triggers	all $J \in [3, 30]$

Values below/above the boundary trigger the social crash; for ς^{th} a lower value triggers, for α_s a higher value triggers.

at the baseline $\alpha_s = 0.60$ with $u^{max} = 1$, which is 34% below the 0.50 cutoff. The analytic boundaries are therefore exact: $\varsigma^{th}_{crit} = 0.33$ and $\alpha^{crit}_s = 0.50/0.55 = 0.909$. In the run that tightens the cutoff toward the boundary, the first sustained exceedance occurs late in the transition (periods 24–55 under the ςth scan), consistent with stress building only as unemployment approaches its peak. Under the pure transfer, 100% of the unemployed stay on relief and ς_t peaks at $\alpha_s = 0.60 > 0.50$, triggering the social channel at the baseline.

Conditional conclusion. The workfare-over-transfer ranking does *not* flip within the scanned range; the sweep strengthens it. To make the recommended corner socially crash, ς^{th} would have to fall from 0.50 to below 0.33 (a 34% reduction) or α_s rise from 0.60 to above 0.91 (a 52% increase), while the pure transfer already socially crashes at the baseline. The persistence count J is immaterial to the corner because stress never exceeds the cutoff at baseline. The ranking remains conditional on the reduced-form cutoffs, whose external empirical anchoring we examine separately (Section 5.3); it is backed by quantified robustness margins rather than a qualitative caveat.

Robustness to the functional form and to feedback. The baseline pressure law is linear in the relief share. Two direct tests ask whether the ranking depends on that choice. The first holds the economic path fixed and replaces the linear conversion with the endpoint-normalized family $g(r) = r^\gamma$, $\varsigma_t = \alpha_s g(r_t)$ with $g(1) = 1$, spanning concave ($\gamma = 0.5$), linear ($\gamma = 1$), and convex ($\gamma = 2$) relief conversion. For the corner and the standard setting the critical α_s that sustains a J-period crash is 0.67, 0.91, and 1.65 across the three exponents, with the linear value 0.91 matching the analytic 0.909; the pure transfer has critical $\alpha_s = 0.50$ under every form because its relief share reaches one, where all conversions

coincide. The second test opens a pressure-to-output channel, $GDP_t \leftarrow GDP_t(1 - \kappa_{fb}\varsigma_t)$ with $\kappa_{fb} \in \{0, 0.25, 0.5, 1.0\}$, which propagates through the capital income and tax base. The corner stays smooth throughout (minimum consumption 0.326 at $\kappa_{fb} = 1$, still above the floor), the standard setting turns economic only at $\kappa_{fb} = 1$ ($\min C = 0.280$), and the pure transfer crashes at every strength. Under both tests and every specification the workfare corner stays smooth while the pure transfer crashes, so the ranking is not an artifact of the linear, no-feedback pressure law. The precise crash boundary is not form-free: it moves from 0.67 to 1.65 across the relief-conversion family, and that dependence is reported rather than absorbed into the wording. Both tests are reproduced by `analysis/check_pressure_form.py` and `analysis/check_pressure_feedback.py`.

5.3. External empirical anchoring of the social-instability parameters

The social-instability channel contains three reduced-form parameters: the social-crash cutoff ς^{th} (baseline 0.50), the stress coefficient α_s (baseline 0.60), and the sustained-exceedance window J (baseline 10 periods, one model period being one year). We search the empirical literature for mappable evidence for each parameter, distinguish direct anchors from indirect (directional) evidence, and state the mapping assumptions explicitly. Table 6 summarizes the anchoring status.

Table 6: External empirical anchoring status of the three social-instability parameters.

Parameter	Baseline	Anchoring status	Nature of evidence
ς^{th}	0.50	no direct external anchor	directional/continuous probability only; no normalized cutoff
α_s	0.60	partially anchored	illustrative reverse-engineered range $\approx$ 1.0–6.6 (model-consistent conversion); baseline 0.60 is below this range, a conservative setting
J	10	partially anchored	mean/upper-tail scale (crisis 8–10 yrs; civil-war mean 11.1 yrs)

The cutoff $\varsigma^{th} = 0.50$: no direct external anchor. No study provides a population-dependency threshold that maps onto a 0.50 cutoff of the $[0, 1]$ normalized stress variable. The authoritative literature consistently reports continuous probability or dose–response relationships rather than structural thresholds replicable across samples. Using rainfall as an instrument across 41 African countries over 1981–1999, Miguel, Satyanath and Sergenti [35] identify that a five-percentage-point drop in real GDP growth raises the probability of conflict in the following year (at least 25 battle deaths) by about 12 percentage points, a relative increase of roughly 50%. Ponticelli and Voth [36], across 25 European countries over 1919–2008, find that a one-standard-deviation spending cut (about 2.8 percentage points of GDP) adds roughly 0.4 unrest events per year, with the event count nearly doubling as cuts widen from 1% to 10% of GDP; only a descriptive soft cutoff around a 2%-of-GDP cut appears, not an identified structural threshold. Using natural disasters as instruments, Bellemare [37] identifies that a one-unit rise in the global food price index adds about 0.99 food-related unrest reports per month, while price volatility has no significant effect. Earlier work by Collier and Hoeffler [47], Fearon and Laitin [48], and Alesina and Perotti [49] likewise reports continuous marginal effects. The only explicit numerical cutoff, due to Lagi, Bertrand and Bar-Yam (2011, arXiv:1108.2455, not peer reviewed), clusters protests when the FAO food price index exceeds 210; but that cutoff is a price-index level rather than a population dependency ratio, and mapping it onto $\varsigma^{th} = 0.50$ would require an over-long chain of normalization assumptions, so we do not adopt it as a direct anchor.

The stress coefficient $\alpha_s = 0.60$: partially anchored, with an identifiable range. The sign with which unemployment or relief dependence raises property crime and social unrest is robust across U.S., European and cross-country panels, ruling out $\alpha_s \leq 0$. The literature reports crime semi-elasticities to unemployment of roughly +1.3% to +4% per percentage point of unemployment, concentrated in property crime with weak effects on violent crime. Raphael and Winter-Ebmer [38] find that a one-percentage-point fall in unemployment lowers property crime by about 1.6%–2.4%; Lin [39] identifies a 1.8% OLS and 4.0% 2SLS rise in property crime per extra percentage point of unemployment; Gould, Weinberg and Mustard [40] find that the 23.3% decline in low-skill male wages over 1979–1993 accounts for 43% of the rise in property crime; Buonanno, Drago and Galbiati [41], across about 20 developed countries

over 1970–2010, identify a roughly 1.3% rise in total crime and 1.7% rise in homicides per extra percentage point of unemployment; and Fougère, Kramarz and Pouget [42] identify a significant positive effect of youth unemployment on property crime in France; see also the surveys by Freeman [50] and Draca and Machin [51]. Mapping these semi-elasticities $\beta_{crime} \approx \partial \log C/\partial u$ onto $\alpha_s = \partial\,\text{stress}/\partial r$ requires three layers of assumptions, and we keep every quantity in consistent units: (A) the normalized range linking the stress index to disorder indicators, setting stress = 1 to correspond to a 3–10-fold amplification of crime over baseline (mapping slopes f' of about 0.43–0.91); (B) the conversion factor $k = r/u$ between the relief share r and unemployment u; and (C) a local linearization treating the semi-elasticity as approximately constant over the working range. On (B), rather than import an external value, we use the model's own accounting: relief recipients are the unemployed not absorbed by public employment, $r = (1 - p_E)u$, so at the baseline $p_E = 0.45$ the conversion is $k = 1 - p_E = 0.55$ (verified directly from the simulated paths). On units, when u is measured on the $[0, 1]$ scale the semi-elasticity entering the chain is $\beta_{crime} = 1.3$–4 (the per-percentage-point 1.3%–4% figures restated per unit of the fraction u). Substituting into $\alpha_s \approx f'\beta_{crime}(1/k)$ with $k = 0.55$ gives an illustrative reverse-engineered interval of roughly 1.0–6.6 (the lower value $0.43 \times 1.3/0.55 \approx 1.02$, the upper $0.91 \times 4/0.55 \approx 6.62$). We read this as a wide, normalization-dependent illustration rather than an empirical estimate: the interval spans more than a factor of six and moves almost one-for-one with the assumed stress-to-disorder mapping. The baseline 0.60 lies *below* even the lower end of this interval, so the calibration deliberately errs on the conservative side, assigning a smaller stress response than the reverse engineering would tolerate; it is a scenario setting rather than a value the interval identifies.

The sustained-exceedance window $J = 10$: partially anchored at mean/upper-tail scale. No study directly estimates the window of consecutive stress exceedance that triggers a crash. Annual-scale evidence brackets ten years along two independent lines. First, after systemic banking crises, per-capita GDP returns to its pre-crisis peak in about eight years on average (median 6.5 years), with 45% of episodes experiencing a second dip [43]; Cerra and Saxena [44] find highly persistent, near-permanent output losses after banking crises, with later estimates placing the average post-crisis GDP loss at about ten years. Second, civil wars last 11.1

years on average (median 7.1 years) [45], and the hazard estimates of Collier, Hoeffler and Söderbom [46] average just over seven years, with wars begun in the 1980s lasting 105 months (about 8.75 years). On the other side, protest waves and crime shocks persist only months to 2–5 years, markedly shorter than ten years. $J = 10$ is therefore a defensible mean/upper-tail anchor aligned with chronic post-crisis slumps and average civil wars, but it lies above the median durations and should be read as such rather than as a median anchor.

Parameter decision and empirical-range robustness. Because none of the three parameters has a precise, directly identified empirical replacement, we keep the baseline values but state their standing accurately: assigning a value to ς^{th} would lack mappable threshold evidence; $\alpha_s = 0.60$ is a conservative setting that sits below the ≈ 1.0–6.6 reverse-engineered interval rather than inside it; and $J = 10$ is a supported mean/upper-tail value. An empirical-range sweep extends the α_s scan across the full 0.30–6.62 interval alongside a directional ς^{th} scan. The analytic boundaries are $\alpha_s^{crit} = 0.50/((1 - 0.45)u^{max}) = 0.909$ (with $u^{max} = 1$) and $\varsigma_{crit}^{th} = 0.33$, and the numerical sweep locates the first social crash at $\alpha_s = 0.91$ (smooth at 0.90) and between $\varsigma^{th} = 0.30$ and 0.33 (smooth at 0.33); $\min C$ stays 0.357 throughout because the social channel does not feed back into the economic paths. The first sustained exceedance occurs late in the transition (periods 24–70 in the α_s sweep). Two results bound the channel. First, the *ranking* is robust across the entire interval: the pure transfer crashes at every α_s (economically at the low end, both economically and socially once $\alpha_s \geq 0.60$), whereas the corner stays smooth through $\alpha_s = 0.90$. Second, the corner's own smoothness is not unconditional: under the upper portion of the reverse-engineered interval ($\alpha_s \geq 0.91$, which includes the ≈ 1.0–6.6 mid-to-high values), the corner also triggers the social channel. The sweep therefore reports both the conservative margin at the baseline and the parameter region in which even the workfare corner would fail, rather than presenting the reduced-form channel as settled.

6. Contribution to the Literature

The analysis belongs to the structural-change tradition that treats technology-driven reallocation of the economy's composition, rather than a single aggregate equilibrium, as the

central object. Pasinetti [61] builds growth on the changing sector composition induced by technical progress, so that structural dynamics precede aggregate analysis. Goodwin [62] casts employment and distribution as an endogenous nonlinear cycle, studied through dynamics rather than comparative statics. Baumol [63] shows how differential productivity growth across sectors produces structural imbalance and a persistent reallocation of labor. The present paper keeps the shared emphasis on technology-driven reallocation and nonlinear dynamics, but locates structural change at the task boundary between humans and AGI rather than at the sector boundary, and studies a case in which the reinstatement engine that reallocates displaced workers collapses endogenously rather than carrying labor through a smooth sectoral shift, with the fiscal and social safety nets closed explicitly. Against that background, the existing AGI macroeconomics literature addresses the transition along three separate axes. Restrepo (2025) and Korinek–Suh (2024) characterize the *supply-side* conditions under which automation collapses wages; Fornaro–Wolf (2026) trace the *demand-side* channel from income shifts to the natural rate; and the fiscal-instrument literature treats taxes and transfers as optimal-schedule objects. Within the studies surveyed here, these axes are not decided together: the size and the financing of the buffer and the existence of the high-employment branch are typically treated as separate questions. This paper proposes a two-layer task model that:

1. **Endogenizes the reinstatement collapse** (driven by AGI participation κ_N^{AI} and the reinstatement rate ω), proving the labor share vanishes monotonically (Theorem 1), and shows skill-retention value decays with ω, with no value as $\omega \to 0$.
2. **Turns the Fornaro–Wolf demand shortfall into a decidable crash channel** (Theorem 4), linking it to the Theorem 1 collapse source.
3. **Characterizes the fiscal closure**: a Laffer interior tax τ^* (Proposition 7) and within-budget superiority of public employment over transfers (Proposition 8).
4. **Introduces a social-instability channel independent by construction, in a separate equation rather than through economic deflation**, proving under the assumed pressure law that workfare dominates the pure transfer for social-instability risk (Proposition 9), complementing the qualitative Tcherneva/Ravallion debate and the basic-income-versus-job-guarantee comparison of Mitchell and Watts [33, 27, 23].

5. **Establishes a robustness ceiling for the buffer family** (Proposition 10): strong shocks require supply-side structural measures, a result absent from the literature.
6. **Closes the open question on the reinstatement rate with an analytic bifurcation theory**. Acemoglu–Restrepo treat κ_N (in their notation) as an exogenous constant and leave its determinants open; we derive the fast subsystem directly from the model and show that the high-employment branch reaches and crosses the switching manifold through a persistence-type boundary equilibrium bifurcation, continuing as a stable low-employment branch on the other side in the frozen fast subsystem. This yields a closed-form critical reinstatement rate, an endogenous policy rule that can carry the effective capture across the time-varying threshold when skill is retained, and a finite-rate lag between the slow policy and fast employment variables (Section 4). The reduction is a local static-slice statement whose parameters are dimensionless and illustrative rather than estimated; on a full path an effective crossing flattens the labor-share decline discontinuously but does not produce interior convergence, so the claim is structural rather than quantitative.

Calibration anchors three magnitudes to data (the consumption floor, the transfer MPC and the workfare wage ratio); the initial labor share $s_0 = 0.90$ follows from the model's own task block, the U.S. labor share of 53.8% is a qualitative benchmark rather than a target, and the elasticity range and the -30%/yr compute-cost decline are background and enter no equation (Section 5). The grid search returns a "workfare-dominant + capital-tax-financed" corner whose coordinates are grid-dependent rather than uniquely identified, providing an analyzable and quantitative basis for a consumption-smooth AGI transition.

Related empirical work grounds the task and labor-share foundations. The skill-content framework originates with Autor, Levy and Murnane [7]; exposure to robots is estimated to reduce local employment and wages [3]; and the declining labor share is linked to the reallocation of value toward superstar firms [6]. Broader policy-oriented analyses emphasize the distributional harms of AI deployed along its current trajectory [1] and the need for development strategies that share the gains from artificial intelligence more widely [21].

Several recent papers address adjacent mechanisms, and in each case the difference lies along a specific axis. Atolia, Holland and Kreamer [5] solve for the majority-voting optimal

capital tax and basic income after an episode of automation, characterizing long-run outcomes and transitional welfare. Their policy is an optimal tax–transfer schedule; ours treats the two instruments as parameters of a bifurcation unfolding and asks when the high-employment branch crosses the switching manifold. Kleven and Zidar [20] characterize optimal labor and capital income taxation in a task-based automation model calibrated to the U.S. wage distribution, and find a sizable optimal capital tax that is, in their benchmark, unaffected by automation. Our capital tax is not an optimal-tax object: it is the financing instrument of the buffer. In the simulated paths capital income $\Pi_t = \max\{0, Y_t - w_t L_t^D\}$ does not shrink as $s_t \to 0$; output growth outpaces the fall in the wage bill and Π_t widens rather than contracts. The binding constraint is therefore the period-by-period timing of revenue rather than a vanishing tax base: with no reserve account, period-t spending is limited to taxes already collected in $t-1$, so early in the transition, before the tax base has grown, planned spending is scaled down. The constraint we identify is this fiscal feasibility rather than tax optimality. Ray and Mookherjee [28] derive the long-run labor share in a growth framework where automation moves the share along a smooth path; we study the case in which the reinstatement engine itself collapses, so the share crosses the switching manifold through a boundary bifurcation rather than following a continuous decline. Nakamura and Zeira [24] analyze transitional unemployment dynamics and identify conditions under which assistance arrives in time; their object is the unemployment stock, ours is the existence of an equilibrium branch. Bond and Kremens [9] characterize income and inequality under asymptotically full automation, a steady-state distribution question, whereas our constraint is fiscal feasibility along the transition. Jacobs [17] studies capitalist–worker wealth distribution in a task-based model without a policy-instrument bifurcation, and Stiefenhofer [31] analyzes AGI substitution inside a CES production structure, where substitution is continuous across factors; our production structure is task-based, so substitution occurs at task boundaries and the resulting instability is discrete. Brault, Haghighi and Yang [16] study monetary policy in a two-sector New Keynesian model with downward nominal wage rigidity and show that accommodating AI-driven unemployment opens an intersectoral inflation wedge because one policy rate cannot clear both labor markets; their analysis is a single-instrument monetary trade-off, whereas our buffer is a fiscal instrument whose financing and the existence of the

high-employment branch are decided together.

7. Conclusion

This paper has built a task-based macroeconomic model of the AGI labor transition that integrates, in one numerically solvable system, the supply side (endogenous reinstatement collapse), the demand side (debt-deflation at the zero lower bound), the fiscal side (capital-tax financing of workfare and transfers), and a social-instability channel. Its results fall into three groups. The *mechanism* group (Theorems 1 and 4) delivers the two failure modes the buffer must contain: a monotonic, non-self-correcting labor-share collapse together with a decidable demand-side spiral. The *fiscal* group (Propositions 6–10) establishes that a financed buffer is nonempty, ranks workfare above transfers within the budget and, under the assumed social-pressure law, against social-instability risk, and bounds the whole family at a robustness ceiling beyond which only supply-side measures help. On the *method*, Section 4 identifies the collapse as a boundary equilibrium bifurcation of the piecewise-smooth fast subsystem, giving a closed-form critical reinstatement rate, and the unified logistic law makes that rate the outcome of competing log-odds: AGI capability drives it down while public employment, workfare and retraining offset the decline, so that a strong, skill-retaining policy can carry the effective capture across the time-varying threshold, durably only in the no-erosion limit, and characterizes the finite-rate lag between slow policy and fast employment.

The policy implications follow within this calibrated model rather than as universal claims. Under the calibrated parameters and at the baseline frontier growth $g_A = 0.010$, the loss-minimizing design on the reported grid is workfare-dominant (a public-employment floor that converts relief recipients into dignified jobs), financed by a minimal sufficient capital tax, with transfers closed because their MPC leakage makes them costlier than workfare. We report the variant that retains a buffer rather than the marginally cheaper one that exhausts it, because the shock test of Section 5 shows that the buffer, not the last hundredth of loss, decides survival. Within this model, the pure transfer and no-policy regimes are infeasible: the floor is left unfunded and the MPC-leakage and social channels turn the path into the modeled social or economic collapse, and Table 1 states the structural conditions under which this result holds. The sufficiency of the design is conditional on the pace of automation: at

$g_A = 0.10$ the consumption floor is breached and the path crashes, so the buffer buys time rather than immunity. For sufficiently fast automation, no demand-side buffer within this model can suffice, so policy must reach the supply side: slowing displacement, strengthening the reinstatement engine, or steering the direction of technical change.

Seven limitations point to future work. First, the social-instability channel is reduced-form; a microfoundation in dignity and acceptability (mapping relief to social pressure) would firm up Proposition 9. Second, the reinstatement rate ω and the automation-frontier growth g_A require sharper empirical calibration (the race between hardware-cost decline and reinstatement). Third, the model is closed-economy and does not model monetary policy beyond the zero lower bound; a monetary-fiscal interaction with active policy, and open-economy capital flows, are natural extensions. Fourth, the core model is deterministic. The frontier-growth shock experiment (Table 3) tests one margin and finds neither configuration crashes at innovations up to $\sigma_\varepsilon = 0.03$, while about a third of the paths of each are right-censored before reaching the employment target, a non-absorption outcome rather than evidence of robustness; the thinner cushion of the refined minimum-loss configuration appears in its fifth-percentile minimum consumption (0.335 versus 0.354 for the headline corner), not in the crash count. Demand-side, monetary and policy-implementation shocks are still absent, and a full stochastic version of the model is left to future work. Fifth, the social-instability parameters still lack strict direct external identification. No study offers a cutoff mappable to the normalized $[0, 1]$ stress variable; α_s is only partially anchored, since crime semi-elasticities, kept in consistent units and combined with the model's own conversion $k = 1 - p_E = 0.55$, reverse-engineer a wide illustrative interval of about 1.0–6.6, with the baseline 0.60 sitting below its lower end as a deliberately conservative setting; and $J = 10$ is a mean/upper-tail anchor drawn from crisis recovery and civil-war durations rather than an estimated exceedance window. The dedicated empirical-range sweep (Section 5.3) bounds the resulting fragility: the workfare-over-transfer ranking holds across the full interval, but the corner remains smooth only through $\alpha_s = 0.90$, turning to a social crash once α_s exceeds 0.91 or ς^{th} falls below 0.33. The baseline values should therefore be read as conservative calibrations not rejected by the available evidence, not as externally estimated quantities, and the empirical samples underlying them (low-income African countries, European welfare states,

and cross-country crisis episodes) differ institutionally from the modeled setting. Sixth, the boundary equilibrium bifurcation is a local analysis of the piecewise-smooth fast subsystem: the endogenous-reinstatement coefficients $\gamma_{PE}, \gamma_{TM}, \gamma_R$, the baseline ω_0 and the capability sensitivity β_A are structural rather than externally estimated, and the policy threshold p_E^* carries the same status as in Section 4.7, a dimensionless output of the calibrated model, illustrative rather than an empirical estimate. The logistic law is given the discrete-choice microfoundation of equation (3), but a quantitative mapping from the analytic critical rate to an estimated time path remains a calibration task left to future work. Seventh, the model is partial-equilibrium rather than a closed national-account system: after-tax capital income and the unconsumed transfer share are recorded as non-returning leakages to capital owners, with no capital-owner consumption, investment or recycling channel, and the bank's lending capacity is an exogenous credit line rather than a deposit-funded balance sheet. Endogenizing capital-owner demand and the funding side of credit would close the flow table fully and, by adding a recycling channel, could only strengthen rather than weaken aggregate demand.

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During the preparation of this work the author used a generative AI assistant in order to improve the language and readability of the manuscript. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

Data availability

The simulation code, parameter files and scripts that generate every reported number are publicly available at `https://github.com/whxhkwj/AGI-transition-workfare-UBI`.

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